



# **MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)**

Multus SDR Control Console for Proficio, Geminus and Ultimus transceivers

Instructions based on MSCC Avalonia

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This preliminary edition covers local operation on an Ubuntu Desktop PC (amd64) or a Raspberry Pi 4/5 (arm64), with the radio plugged into that computer, plus firmware updates (Appendix B) and WSJT-X setup (Appendix C).

Remote operation has its own guide (coming). A separate guide covers MSCC for Windows.

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## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

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## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

### □ WELCOME

This guide covers installing and using MSCC, the Multus SDR Control Console, on a **Linux PC running Ubuntu Desktop (64-bit, “amd64”)** or on a **Raspberry Pi 4 or 5 running 64-bit Raspberry Pi OS (“arm64”)**, with the radio plugged into that same computer. That is called *local operation*. MSCC is the control program for Multus SDR, LLC transceivers: Proficio, Geminus and Ultimus, in both Legacy and MKII versions. It works only with Multus SDR equipment.

MSCC comes in two parts:

- **The servers.** These small programs talk to the radio over USB and do the receive and transmit signal processing. They run on the computer the radio is plugged into. Their names are `sdrcore-recv`, `sdrcore-trans` and `ms-sdr`; you will see these names in some messages.
- **The client.** This is the window you operate from. On Linux and the Raspberry Pi it is called **MSCC UI** in the menu, and its window title reads **MSCC Avalonia**. These instructions are based on MSCC Avalonia.

A setup program, **MSCC Init**, tells the servers which radio, serial port, speaker and microphone to use. You run it once after installing, and again whenever you change your audio devices.

The client and servers talk over your network connection, even when they are on the same computer. Operating from another computer in the house or across the internet (“remote”) has its own guide and is not covered here.



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MSCC UI connected to the radio, MAIN tab showing



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### □ 1. Overview (READ THIS FIRST!)

Please read this section before you install anything.

1. **Firmware first.** The radio's firmware should match this version of MSCC. The window title shows the firmware version (FW) after you connect. Firmware updates are covered in Appendix B.
2. **Which computer.** This guide is for a Linux PC with Ubuntu Desktop, or a Raspberry Pi 4 or 5. The two use different install files (Section 2.2), but once installed, MSCC looks and works the same on both. Pictures in this guide were taken on an Ubuntu PC.
3. **Set up once with MSCC Init.** After installing, run MSCC Init (Section 2.4). MSCC UI will warn you if it hasn't been done.
4. **Calibrate once.** A new radio or new installation should get frequency calibration (FREQ CAL), power calibration (QRP CAL) and transmit image balance (TX IQ). Factory values are loaded automatically for your radio model, so this is a touch-up, not a must-do on day one.
5. **Version numbers in this guide are examples.** File names such as `mscc_1.0.xx_amd64.deb` or `mscc-ui_0.6.xx_amd64.deb` stand for whatever version you were given. Always use the newest files you have; you don't need to look for the exact version shown here.
6. **Bug reports and feature requests:** post them on our forum, <https://multusdr.groups.io/g/main>



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# □ 2. Installing and First Run (Instructions based on MSCC Avalonia)

## 2.1 What you need

- A **Linux PC** with **Ubuntu Desktop, 64-bit** (the “amd64” kit), or a **Raspberry Pi 4 or 5** with **64-bit Raspberry Pi OS**, desktop version (the “arm64” kit).
- The radio, with its USB cable plugged into that computer.
- A speaker or headphones and a microphone on that computer (the computer’s own audio, a USB headset, or a Pi audio HAT).
- The MSCC install files for your computer, in one folder. The **Ubuntu kit** has names ending in `_amd64.deb`; the **Raspberry Pi kit** has names ending in `_arm64.deb`. Both kits also contain `install-mscc.sh` and an `mscc-init-gui_1.0.xx_all.deb` file.
- A mouse with a scroll wheel. MSCC uses the wheel for tuning.

**Use the right kit.** Never install `_arm64` packages on an Ubuntu PC, or `_amd64` packages on a Raspberry Pi. The only file that works on both is `mscc-init-gui_1.0.xx_all.deb`.

## 2.2 Installing the packages

You install MSCC by typing (or pasting) a few commands into a **Terminal** window. Each command below can be copied and pasted as it is. When a command starts with `sudo`, the computer asks for your login password; typing it shows nothing on the screen, which is normal. Press Enter after each command and wait for it to finish.

**Open a Terminal in the folder with the install files.** In the Files app, open the folder, right-click an empty spot and choose **Open in Terminal**. (Or open Terminal and type `cd` followed by a space, drag the folder into the Terminal window, and press Enter.)

### Ubuntu Desktop PC (amd64 kit)

1. Give your login permission to use the radio’s serial and audio ports:

```
sudo usermod -aG dialout,audio,plugdev "$USER"
```

2. **Log out and log back in** (or restart the computer). This step is needed for the permission to take effect. Then open the Terminal in the install folder again.
3. Install the four packages, **in this order** (audio library, servers, MSCC Init, MSCC UI):



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```
sudo apt install -y ./mscc-portaudio_*_amd64.deb
chmod +x install-mscc.sh
./install-mscc.sh
sudo apt install -y ./mscc-init-gui_*_all.deb
sudo apt install -y ./mscc-ui_*_amd64.deb
```

The \* in each name means “whatever version is in the folder”, so you don’t need to type the version number. `install-mscc.sh` installs the servers package (for example `mscc_1.0.xx_amd64.deb`); run it as yourself, not with `sudo`. It asks for your password when it needs it.

**Don’t install “pulseaudio” on Ubuntu.** Ubuntu already has its own sound system (PipeWire), and MSCC works with it.

### Raspberry Pi 4 or 5 (arm64 kit)

1. Install the packages, **in this order**:

```
sudo apt install -y ./mscc-portaudio_*_arm64.deb
chmod +x install-mscc.sh
./install-mscc.sh
sudo apt install -y ./mscc-init-gui_*_all.deb
sudo apt install -y ./mscc-ui_*_arm64.deb
```

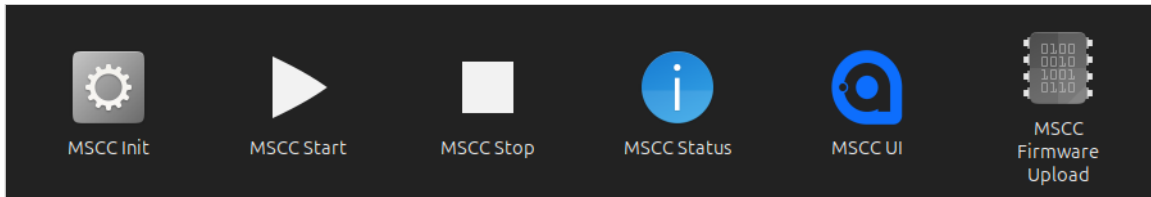
2. **Log out and log back in** (or restart the Pi). This gives your login access to the serial port and audio.

The Pi kit may also contain `proficio-flash-tools_1.0.xx_all.deb`. It is optional and not needed for normal operation. **Question for Ron:** should Pi users also run `sudo usermod -aG dialout,audio,plugdev &quot;$USER&quot;` (as in the Ubuntu steps) before logging out, or does the Pi servers package do this?

## 2.3 The MSCC menu

After installing, six MSCC entries appear with your other programs. On Ubuntu, click the grid of dots (**Show Apps**) at the bottom of the dock. On the Raspberry Pi, open the start menu (the Raspberry icon at the top left of the screen): it has an **MSCC** folder, and the MSCC entries are inside it. **Screenshot coming:** picture of the Raspberry Pi OS start menu with the MSCC folder open

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*The MSCC entries in the Ubuntu app grid: MSCC Init, MSCC Start, MSCC Stop, MSCC Status, MSCC UI, and MSCC Firmware Upload*

| Menu entry                  | What it does                                                                           | See        |
|-----------------------------|----------------------------------------------------------------------------------------|------------|
| <b>MSCC Init</b>            | Sets up MSCC for this computer: radio, keyer, CAT serial port, speaker and microphone. | 2.4        |
| <b>MSCC Start</b>           | Starts the servers.                                                                    | 3.4        |
| <b>MSCC Stop</b>            | Stops the servers.                                                                     | 3.4        |
| <b>MSCC Status</b>          | Opens a Terminal window with a health check of the servers and settings.               | 3.4        |
| <b>MSCC UI</b>              | The operating window (MSCC Avalonia).                                                  | 3, 4       |
| <b>MSCC Firmware Upload</b> | Loads new firmware into the radio.                                                     | Appendix B |

On Ubuntu you can right-click an entry and choose **Pin to Dash** to keep it on the dock.

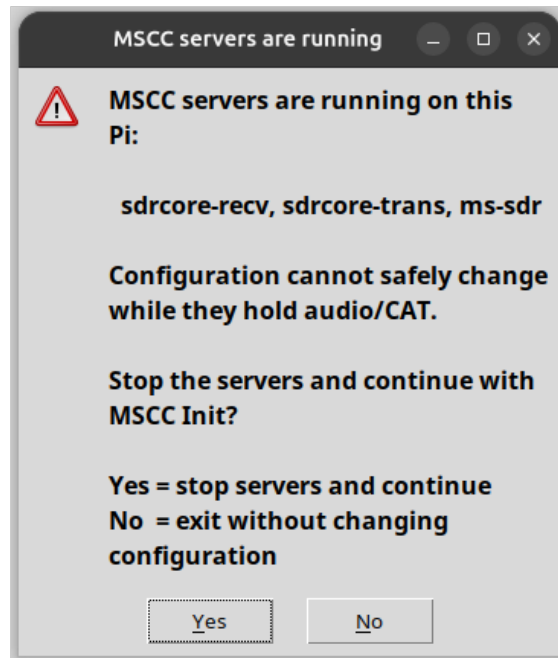
### 2.4 MSCC Init: setting up this computer

Run **MSCC Init** once after installing, and again if you change your speaker, microphone or radio. It has four steps, then a review page. Use **Next** → and ← **Back** to move between the steps; **Cancel** leaves without changing anything.

**The heading says “configure this Pi” even on an Ubuntu PC.** That’s normal; MSCC Init is the same program on both.

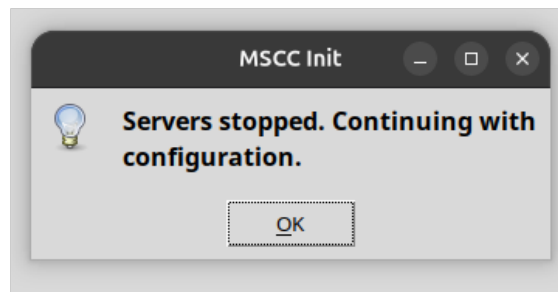
If the servers are running when you open MSCC Init, it asks to stop them first:

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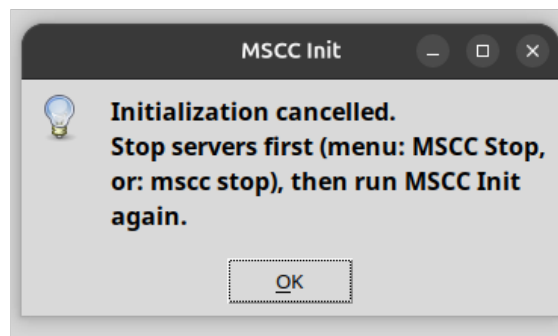


*MSCC Init: "MSCC servers are running on this Pi". Yes stops them and continues; No exits without changing anything*

- **Yes** stops the servers and continues. You'll see "Servers stopped. Continuing with configuration." Click **OK**.
- **No** closes MSCC Init without changing anything ("Initialization cancelled. Stop servers first (menu: MSCC Stop, or: mscs stop), then run MSCC Init again.").



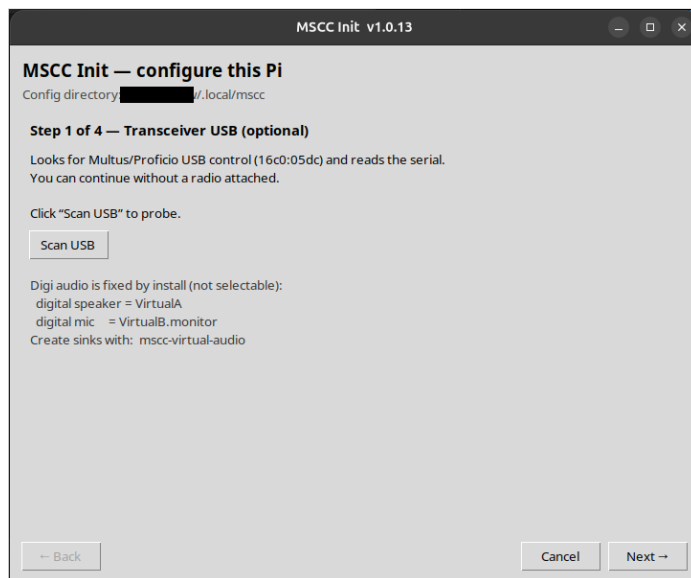
*After Yes: "Servers stopped. Continuing with configuration."*



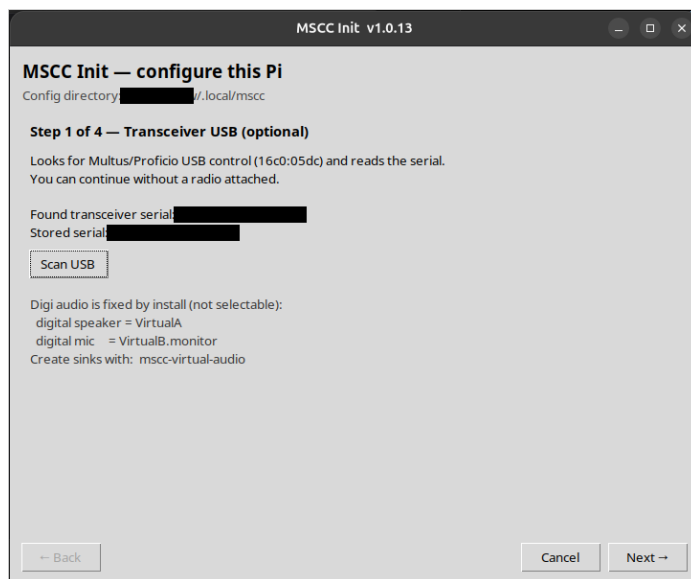
*After No: "Initialization cancelled."*

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**Step 1 of 4: Transceiver USB (optional).** Click **Scan USB** with the radio plugged in and powered on. MSCC Init finds the radio and shows its serial number (“Found transceiver serial”), and the one already saved (“Stored serial”). You can continue without a radio attached. The page also shows the fixed digital audio devices (**VirtualA** and **VirtualB.monitor**); these are created by MSCC and are not something you pick. Click **Next** →.



*Step 1 before Scan USB*



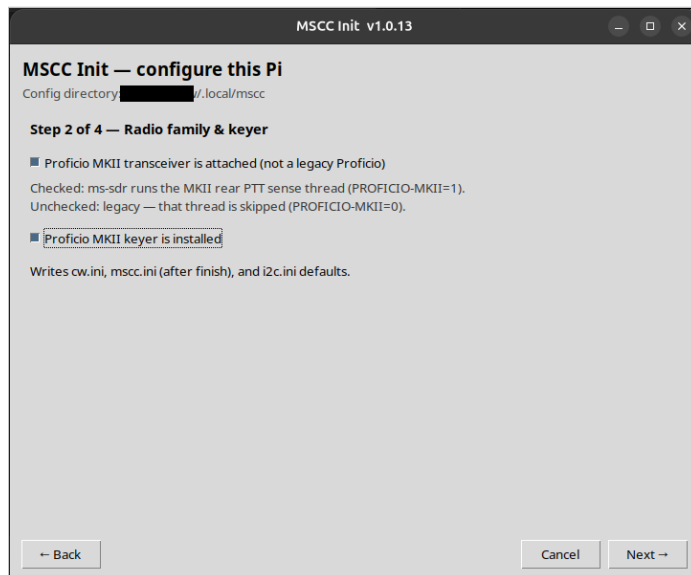
*Step 1 after Scan USB: the radio's serial number is shown (blacked out here)*

**Step 2 of 4: Radio family and keyer.**

- **Proficio MKII transceiver is attached (not a legacy Proficio).** Check it for an MKII radio. Uncheck it for a Legacy radio.

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- **Proficio MKII keyer is installed.** Check it to use the radio's built-in electronic keyer. Uncheck it if you use your own external keyer (see also the CW tab, 4.2).

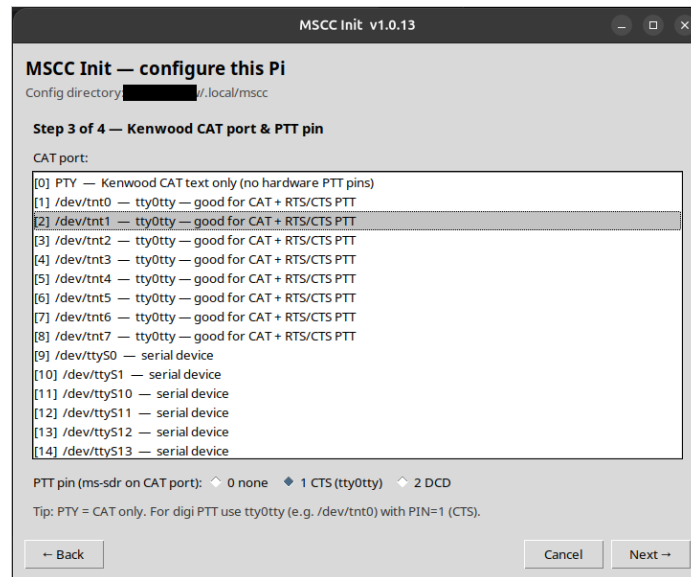


*Step 2: radio family and keyer*

**Step 3 of 4: Kenwood CAT port and PTT pin.** This is the serial port the servers use for CAT, the connection your logging or digital program (WSJT-X) uses to set frequency and key the transmitter.

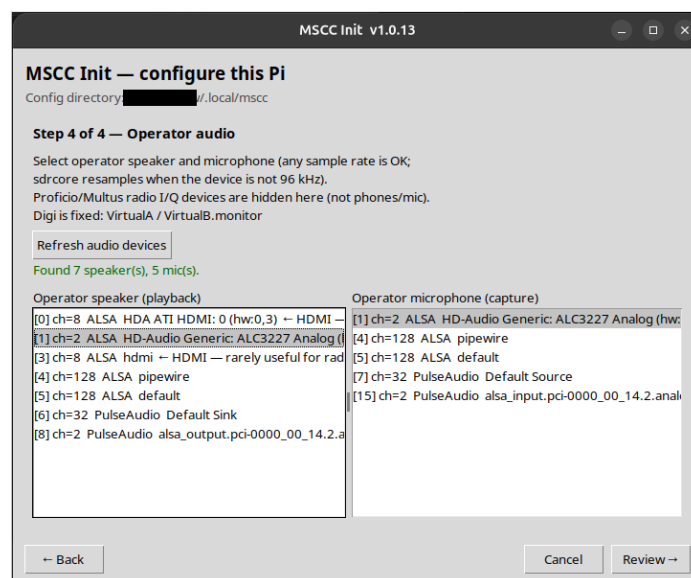
- Pick one of the `/dev/tnt` ports (“`tty0tty` — good for CAT + RTS/CTS PTT”). The port depends on what you pick here. These virtual serial ports (made by MSCC’s `tty0tty` driver) come in pairs, `tnt0` with `tnt1`: the servers use the end you pick, and your digital program (WSJT-X etc.) uses the other end of the same pair. In the example the servers use `/dev/tnt1`, so WSJT-X uses `/dev/tnt0` (Appendix C). Always check SETTINGS in MSCC UI (4.10), which shows which end is which.
- **PTT pin:** choose **1 CTS (`tty0tty`)**.

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Step 3: CAT port /dev/tnt1 and PTT pin 1 CTS selected

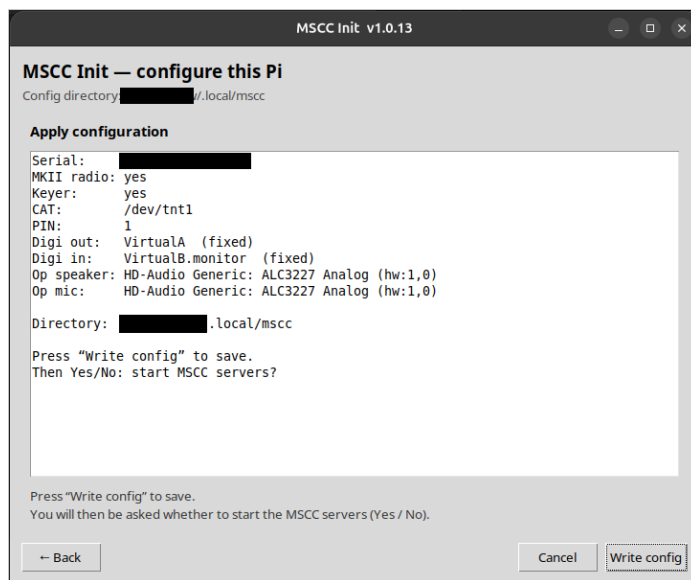
**Step 4 of 4: Operator audio.** Pick the **Operator speaker (playback)** and the **Operator microphone (capture)** you want to use for voice and for hearing the radio. Any sample rate is fine. The radio's own USB audio and the digital devices are hidden from these lists on purpose. Click **Refresh audio devices** if you plugged in a headset after opening MSCC Init. On a PC, choose the computer's built-in audio (in the example "HD-Audio Generic: ALC3227 Analog") or your headset, not an HDMI output (HDMI is usually your monitor). Click **Review** →.



Step 4: operator speaker and microphone

**Apply configuration.** The review page lists your choices. Check them, then click **Write config**.

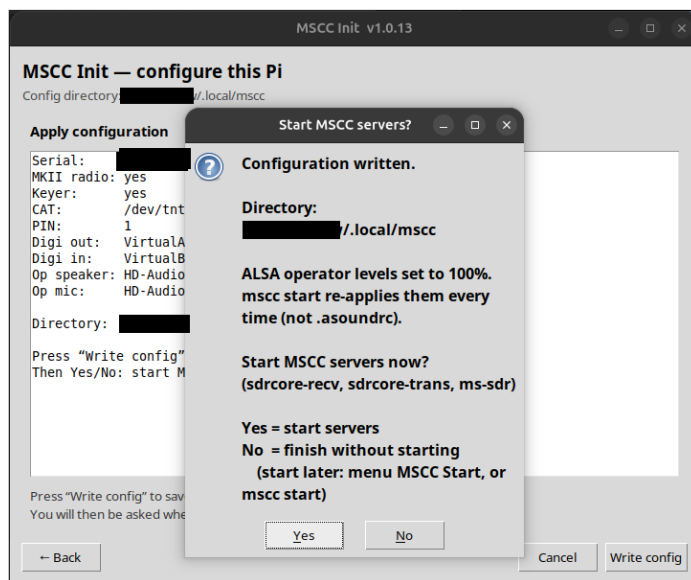
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*Review page: click Write config to save*

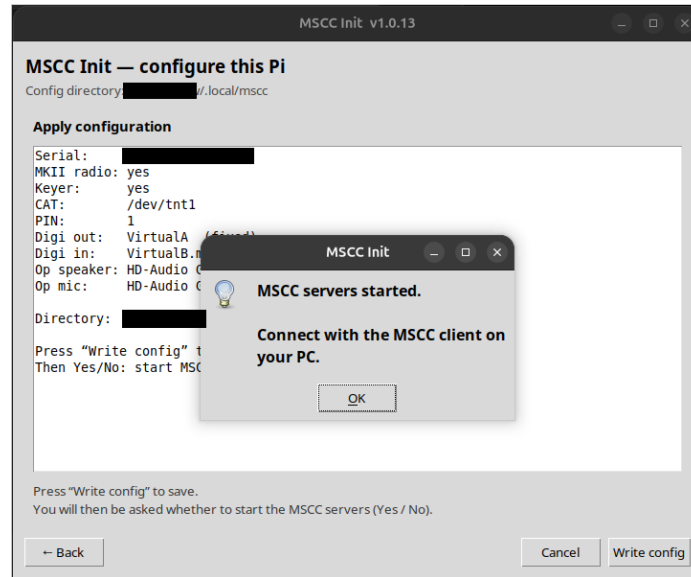
MSCC Init saves the settings and asks **“Start MSCC servers now?”**

- **Yes** starts the servers. You’ll see **“MSCC servers started.”** Click **OK**. (The message says “Connect with the MSCC client on your PC”; on a local setup, that means MSCC UI on this same computer.)
- **No** finishes without starting them. Start them later with **MSCC Start**, or let MSCC UI start them for you (3.1).



*Start MSCC servers now? Yes = start servers, No = finish without starting*

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*MSCC servers started*

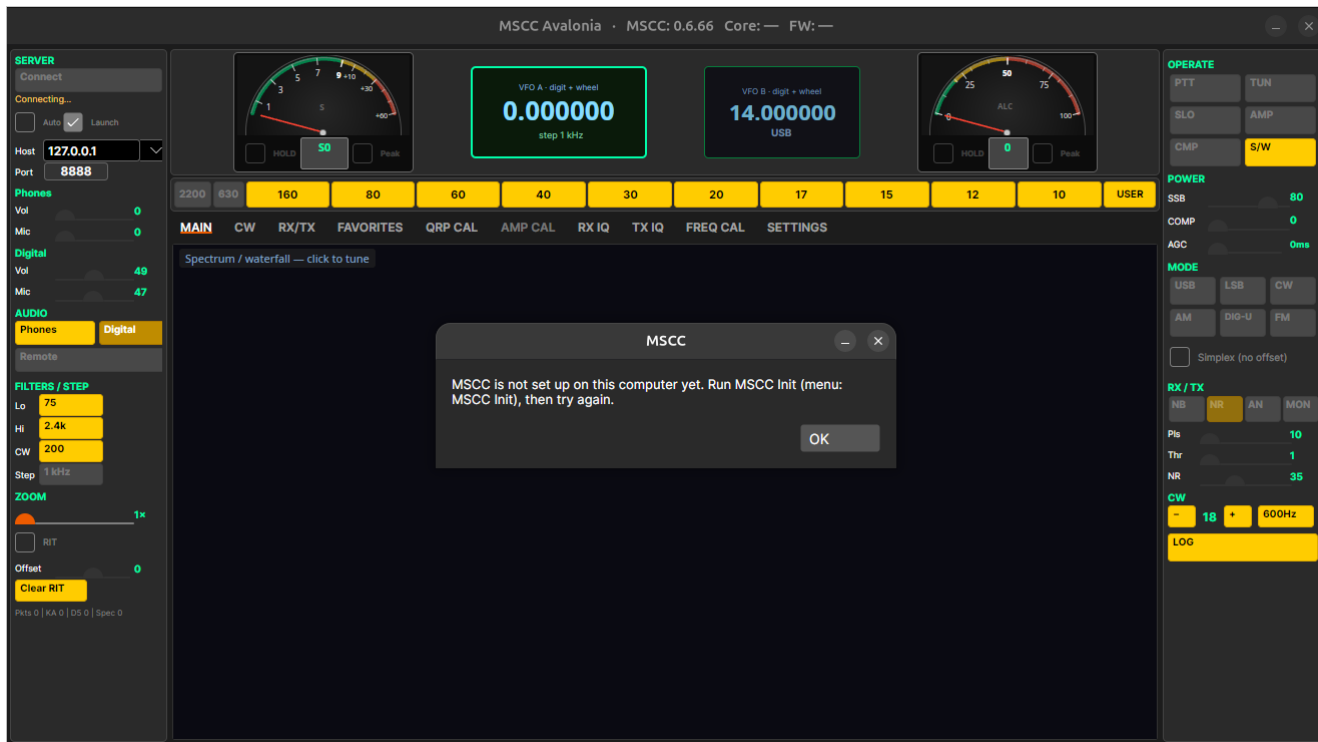
Close MSCC Init when you're done.

### 2.5 Starting MSCC UI for the first time

1. Plug in the radio's USB cable and power the radio on.
2. Start **MSCC UI** from the MSCC menu.
3. For local operation leave **Host** at **127.0.0.1** and **Port** at **8888**.
4. Check **Launch** (Section 3.1) so MSCC UI starts the servers for you, or start them first with **MSCC Start**.
5. Press **Connect**.

**"MSCC is not set up on this computer yet."** If MSCC Init hasn't been run (or its files are missing), pressing Connect with Launch checked shows this message:

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*Warning shown on Connect when MSCC Init has not been run*

This is a **warning**, not an error. Click **OK** and MSCC UI carries on and tries to start the servers and connect. If the connection works, you can keep operating, but the CAT port and your speaker and microphone may not be what you want. Run **MSCC Init** (2.4) when you can, then Connect again.

Only one copy of MSCC UI should be open at a time. The window remembers its size and position, and your frequencies, modes, filters and display settings are remembered between sessions.

## 2.6 Upgrading

1. Close MSCC UI, or press Disconnect.
2. Install the new files the same way as in 2.2 (the same commands upgrade in place). You only need to install the packages that changed. On Ubuntu you don't need to repeat the usermod command or log out again.
3. Start MSCC UI and Connect. If the servers were already running, stop and start them once (MSCC Stop, then MSCC Start) so the new version is used.

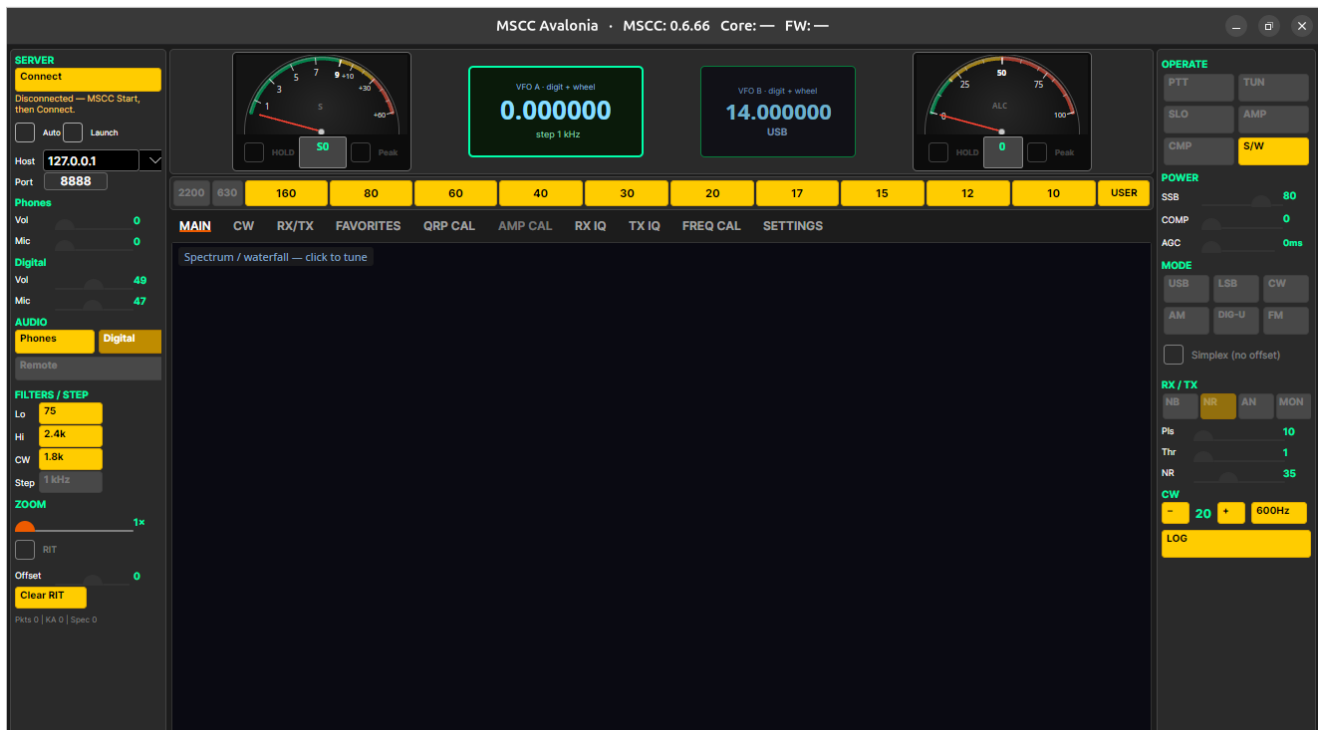
Your settings are kept when you upgrade: MSCC UI's settings in `~/.config/MSCC/` and the server settings (from MSCC Init) and calibration in `~/.local/mscc/`. You don't need to run MSCC Init again unless your audio devices changed. **Question for Ron:** does a servers-package upgrade ever overwrite the calibration files in `~/.local/mscc/`?

### 3. Starting, Connecting and Stopping

#### 3.1 Start options: Auto and Launch

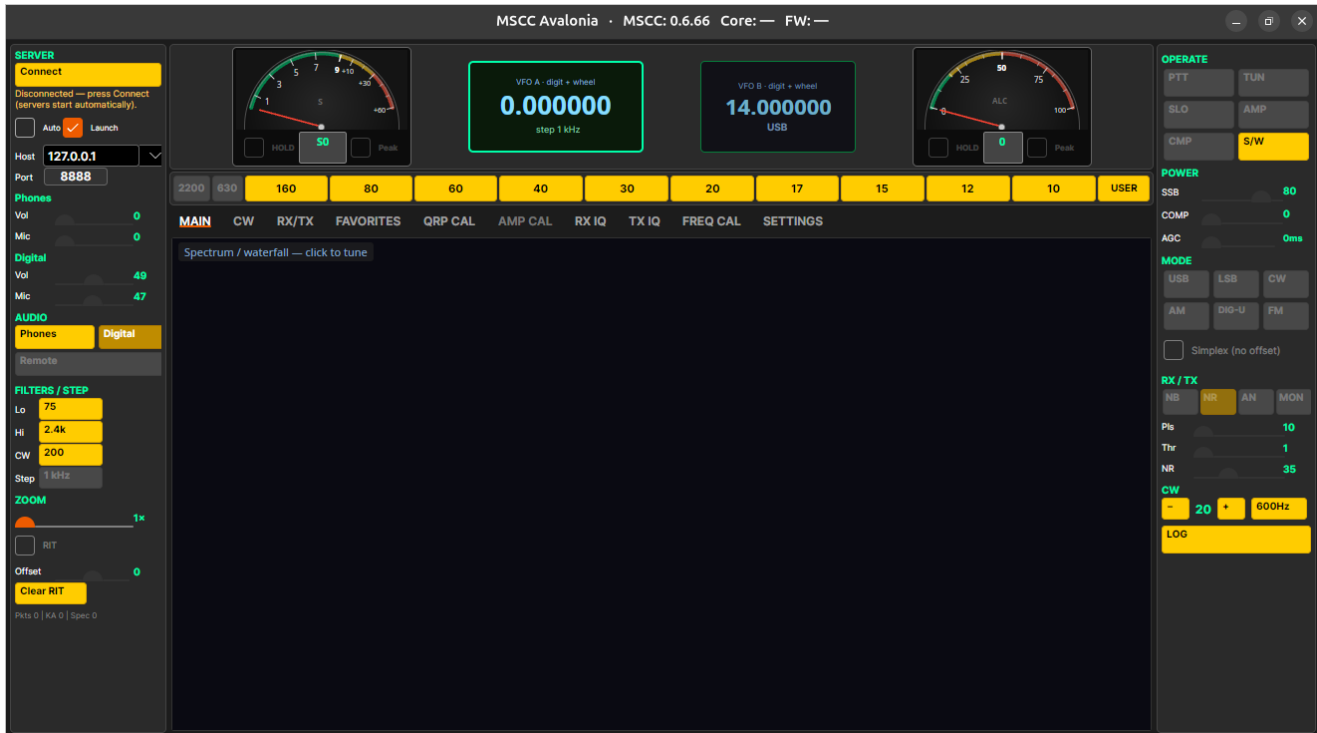
On the left rail, under **SERVER**:

- **Connect / Disconnect.** Connects to the servers. While connected it reads **Disconnect**.
- **Launch** checkbox (local operation only). When checked, **Connect** also starts the servers on this computer if they aren't already running (the same as MSCC Start), and **Disconnect** stops them again. Launch can be changed only while disconnected, and only when Host is **127.0.0.1**; for any other Host it is grayed out.
- **Auto** checkbox. Presses Connect for you as soon as MSCC UI opens. The servers are started only if Launch is also checked.
- **Host.** The address of the servers. For local operation this is always **127.0.0.1** ("this computer"). A change takes effect on the next Connect. The ▼ button lists addresses you used before.
- **Port.** **8888** unless it was changed on the servers.

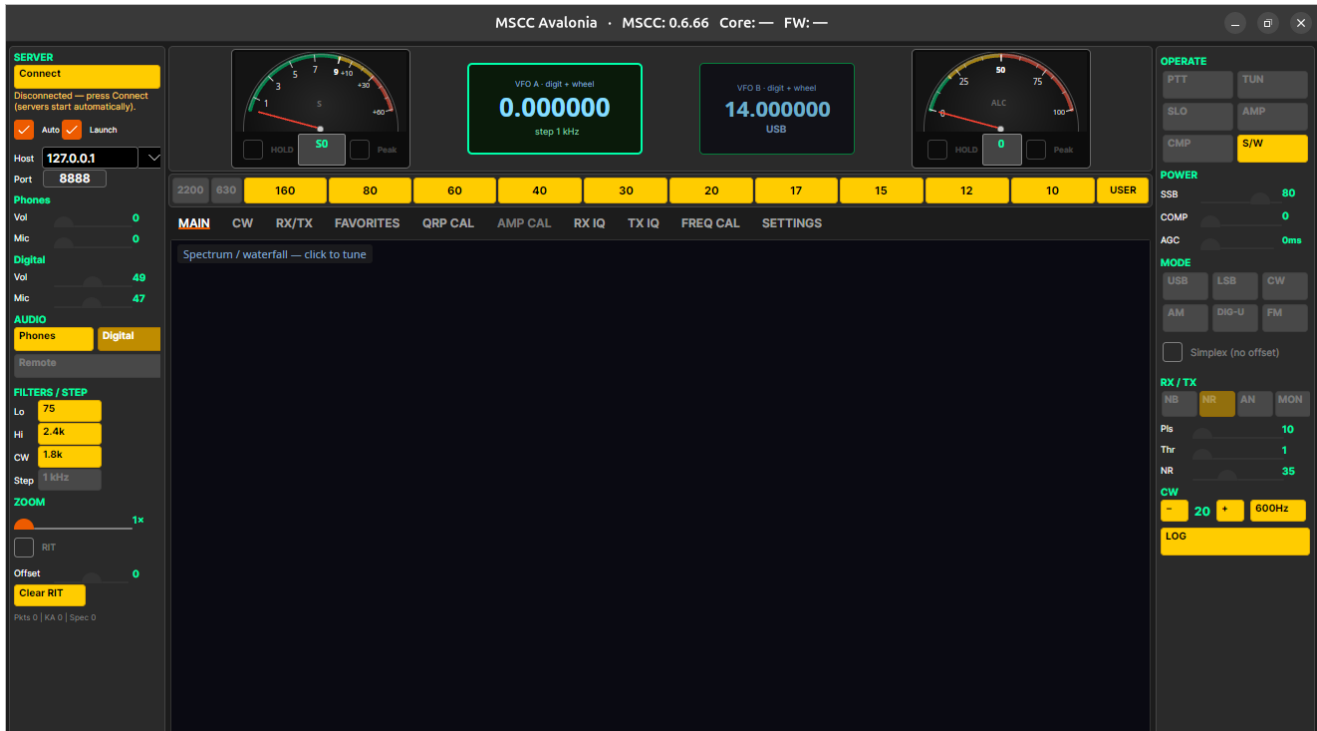


*Launch unchecked: start the servers first with MSCC Start, then Connect*

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*Launch checked: Connect starts the servers automatically*



*Auto and Launch both checked: MSCC UI starts the servers and connects by itself when it opens*

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The line under the Connect button tells you what Connect will do:

- **“Disconnected — MSCC Start, then Connect.”** Launch is off. Start the servers with **MSCC Start** first, then press **Connect**.
- **“Disconnected — press Connect (servers start automatically).”** Launch is on.

**Which to use?** For most local stations, check **Launch** (and **Auto** if you like). MSCC UI then starts and stops everything for you. If you prefer to start the servers yourself (for example you also run WSJT-X without MSCC UI), leave **Launch** unchecked and use **MSCC Start** and **MSCC Stop**.

**If the servers are already running** (for example you used **MSCC Start**), **Connect** just connects, even with **Launch** checked, and **Disconnect** leaves those servers running. MSCC UI only stops servers that it started itself.

### 3.2 What you see while connecting

While the servers start, the status line reads **“Starting servers...”** and then **“Waiting for servers...”**. This takes several seconds. Ubuntu also shows a desktop notification, **“Servers started (recv, trans, ms-sdr).”**, which fades away by itself.



*Starting servers...*



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*Desktop notification: Servers started (recv, trans, ms-sdr)*

When the connection is working, the button reads **Disconnect**, the status line reads **“Connected — spectrum OK”**, the VFOs show your frequency, and the spectrum and waterfall come alive. The title bar then shows the versions and your radio model. For example (the numbers on yours will differ): MSCC Avalonia · MSCC: 0.6.xx Core: 3.171 FW: 3.232 Proficio MKII PTT

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*Connected, with spectrum and waterfall*

If the status line shows **“WARNING: keep-alive lost”**, MSCC UI has stopped hearing from the servers. Check that the radio is on and its USB cable is plugged in, then Disconnect and Connect again. If that doesn’t help, see Appendix E.

### 3.3 Disconnecting and closing

**Disconnect** turns transmit off, ends any calibration that is running, and ends the session. If MSCC UI started the servers (Launch), it also stops them, and Ubuntu shows **“Servers stopped.”** Closing the MSCC UI window does the same.

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*Desktop notification after Disconnect: Servers stopped*

### 3.4 MSCC Start, MSCC Stop and MSCC Status

These menu entries control the servers directly. You don't need them if you use **Launch** in MSCC UI, but they are handy for checking and fixing things.

- **MSCC Start** starts the servers and shows “**Servers started (recv, trans, ms-sdr).**” If it fails you'll see “Start failed. Try Terminal: msc start”.
- **MSCC Stop** stops the servers and shows “**Servers stopped.**” Stop the servers before you run MSCC Init or update the firmware.
- **MSCC Status** opens a Terminal window with a full report: whether each server is running, the settings MSCC Init wrote, whether your audio devices were found, and whether the CAT port works. Lines start with **OK**, **WARN** or **FAIL**. The **Summary** at the end should read “**MSCC status: OK**” and “**FAIL=0 WARN=0**”. If you ask for help, a picture of this window is very useful. Close the window when you're done.



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```
mscc/mscc-status-window.sh
MSCC status - [redacted] mscc
sdrcore-recv: running (pid 14584)
sdrcore-trans: running (pid 14606)
ms-sdr: running (pid 14627)
MSCC status report - [redacted]

=== Servers ===
OK sdrcore-recv: running (pid 14584)
OK sdrcore-trans: running (pid 14606)
OK ms-sdr: running (pid 14627)

=== Binaries ===
MSCC_DIR [redacted] mscc
OK sdrcore-recv
OK sdrcore-trans
OK ms-sdr
OK mscc.sh

=== PortAudio (library used by cores) ===
libportaudio.so.2 => /usr/local/lib/libportaudio.so.2 (0x00007e15aeb6a000)
OK linked against /usr/local (mscc-portaudio)

=== Config ($HOME/.local/mscc) ===
config dir: [redacted].local/mscc
OK operator speaker: HD-Audio Generic: ALC3227 Analog
OK operator mic: HD-Audio Generic: ALC3227 Analog
OK digital speaker: VirtualA
OK digital mic: VirtualB.monitor
OK CAT / comm: COMM PORT_NAME=/dev/tty1 PIN=1
file: [redacted]local/mscc/comm-port.ini
OK mscc.ini: serial: [redacted] MSCC_IP: [redacted] ports 8889/8888
```

*MSCC Status, top of the report*

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```
mscc/mscc-status-window.sh

=== PortAudio devices vs config ===
PortAudio sees 21 devices; host APIs: ALSA, PulseAudio
OK Proficio/Multus present: pa=8 api=PulseAudio in=0 out=2 name='alsa_outpu
t.usb-www.multus-sdr .01.analog-stereo'
OK Proficio/Multus present: pa=15 api=PulseAudio in=2 out=0 name='alsa_outp
ut.usb-www.multus-sdr 01.analog-stereo.monitor'
OK Proficio/Multus present: pa=16 api=PulseAudio in=2 out=0 name='alsa_inpu
t.usb-www.multus-sdr .01.analog-stereo'
OK Operator speaker: matched pa=1 api=ALSA in=2 out=2 rate=44100 name='HD-A
udio Generic: ALC3227 Analog (hw:1,0)'
OK Operator mic: matched pa=1 api=ALSA in=2 out=2 rate=44100 name='HD-Audio
Generic: ALC3227 Analog (hw:1,0)'
OK Digital speaker: matched pa=9 api=PulseAudio in=0 out=2 rate=48000 name=
'VirtualA'
OK Digital mic: matched pa=18 api=PulseAudio in=2 out=0 rate=48000 name='Vi
rtualB.monitor'

=== Digital sinks (Pulse/PipeWire) ===
OK sink VirtualA present
OK sink VirtualB present

=== CAT / serial ===
OK device exists: /dev/tnt1 PIN=1
mode=00660
OK read/write access OK for /dev/tnt1

=== Proficio USB (control 16c0:05dc) ===
OK USB control device present serial:

=== Summary ===
MSCC status: OK
FAIL=0 WARN=0 (exit 0)
```

*MSCC Status, end of the report: Summary OK, FAIL=0 WARN=0*

The same commands can be typed in a Terminal:

```
mscc start
mscc stop
mscc status
```

`mscc restart` stops and starts the servers in one go.

### □ 4. Local Operation (Instructions based on MSCC Avalonia)

#### 4.1 The Main Window



*The main window, connected (MAIN tab)*

The main window has three columns:

- **Left rail:** connection, audio, filters, zoom and RIT.
- **Center:** meters and VFOs at the top, the band buttons, then the tabs.
- **Right rail:** the operating controls (PTT, TUN, power, modes, noise controls, CW).

Both rails stay visible on every tab.

#### Button colors:

- Bright gold = off.
- Darker gold (brown) = on or selected.
- **Red** = transmitting, or a transmit-related switch that is on (PTT, TUN, calibration TX).
- Gray = not available right now (for example most controls before you connect).

**Mouse wheel:** over any slider, one notch moves the slider one small step.



## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

### Left rail

1. **Connect / Disconnect** (SERVER), with a status line under it. See 3.1.
2. **Auto** and **Launch** checkboxes. See 3.1.
3. **Host** and the recent-hosts ▼ button. **127.0.0.1** for local operation.
4. **Port**. Normally 8888.
5. **Phones Vol** (0–100). The headphone and speaker volume.
6. **Phones Mic** (0–100). The microphone gain.
7. **Digital Vol** (0–100). Receive audio level sent to your digital program (WSJT-X).
8. **Digital Mic** (0–100). Transmit audio level coming from your digital program.
9. **Phones** button (AUDIO). Use your speaker or headphones and microphone.
10. **Digital** button (AUDIO). Use the digital audio path for WSJT-X and similar programs. Choosing the **DIG-U** mode switches to Digital for you; leaving DIG-U puts back what you had.
11. **Remote** button. Grayed out for local operation (Host 127.0.0.1). It is used only when the radio is on another computer; see the Remote guide.
12. **Lo** (low cut). Each click moves the low edge of the receive filter: 500 → 300 → 200 → 100 → 75 Hz, then back to 500.
13. **Hi** (high cut). Each click moves the high edge of the receive filter.
  - In **DIG-U**: 5.5k → 4.0k → 3.0k → 2.7k → 2.4k → 1.4k → 1.0k, then back to 5.5k. **The 1.4k and 1.0k settings are for DIG-U only.** They are handy for narrow digital work.
  - In **all other modes**: 5.5k → 4.0k → 3.0k → 2.7k → 2.4k.
14. **CW** (CW bandwidth). 1.8 kHz → 400 Hz → 200 Hz. Used in CW mode.
15. **Step**. Tuning step for the mouse wheel: 10 Hz → 100 Hz → 1 kHz → 10 kHz → 100 kHz. You can also double-click VFO A to change it.
16. **ZOOM** (1×–4×). Magnifies the spectrum and waterfall around your frequency.
17. **RIT** checkbox. Turns receive offset on.
18. **Offset** slider (–500 to +500 Hz). The RIT offset.
19. **Clear RIT**. Turns RIT off and sets the offset back to 0.
20. **Counters** (small gray text at the bottom, “Pkts ... | KA ... | DS ... | Spec ...”). Show that data is arriving from the servers. Useful when reporting a problem; otherwise you can ignore them.

### Meters and VFOs (top of the center column)

21. **S-meter**. Received signal strength. The reading box under it shows the S reading. The **HOLD** and **Peak** boxes are shown but don't do anything yet.

## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

22. **VFO A.** Click it to make VFO A active; the active VFO has a bright border. To tune:
- Put the mouse pointer over a digit and roll the wheel. That digit changes, and the digits to its right go to zero. Over the 1 kHz digit, for example, you tune in 1 kHz steps.
  - Roll the wheel over the VFO box but not over a digit to tune by the **Step** value (shown under the mode, for example “step 1 kHz”).
  - **Double-click** VFO A to change the Step.
  - Under the frequency: the current mode (USB, LSB, CW, AM, DIG-U, FM, TUNE).
23. **VFO B.** Works the same way as VFO A. Each VFO remembers its own band. In FM repeater mode, VFO B shows the transmit frequency.
24. **ALC meter.** Transmit ALC level. It rests at zero on receive. HOLD and Peak don't do anything yet.

### Band buttons

25. **Band buttons: 2200, 630, 160, 80, 60, 40, 30, 20, 17, 15, 12, 10 and USER.** Click a band to go there. Bands your radio can't use are grayed out: Proficio and Ultimus are HF radios (160–10 m), and Geminus is an LF/MF radio (2200 m and 630 m).
- Each band button returns to the frequency, mode and filters you last used on that band.
  - The active band button is darker.
  - **USER** (general coverage and time stations). Each click steps through WWV 5, 10, 15 and 20 MHz, CHU 3.330 and 7.850 MHz, RWM 9.996 MHz, and USER (your own last general-coverage frequency). These stations are good references for FREQ CAL.

### Tabs

26. **Tab strip:** MAIN, CW, RX/TX, FAVORITES, QRP CAL, AMP CAL, RX IQ, TX IQ, FREQ CAL, SETTINGS.
- QRP CAL and TX IQ are for AMP off; AMP CAL is for AMP on. The tab that doesn't match the AMP setting is shown in gray.
  - When you leave FREQ CAL, MSCC puts back the mode, filter and pitch you had before.

### MAIN tab

27. **Spectrum and waterfall.**
- **Left-click** a signal to tune to it.
  - The line at the top shows the center frequency, the display resolution, the zoom and “click to tune”.
  - The shaded band on the spectrum is your receive passband.
  - Colors and levels are set in the S/W window (4.11).



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### Right rail

- 28. **PTT.** Transmit on/off; red while on. For voice modes (USB, LSB, AM, FM). In CW, the radio is keyed by the paddle or keyer, not by PTT. PTT and TUN are locked while a digital program or hardware PTT has control of transmit ("Server owns TX — PTT/TUN locked").
- 29. **TUN.** Sends a steady carrier at **Tune Power** for your tuner or amplifier; red while on. Click again to return to your previous mode.
- 30. **SLO / MED / FST** (AGC). Each click changes the AGC speed.
- 31. **AMP.** Switches between the radio's two power tables so you can go from barefoot (QRP) to an external amplifier and back without overdriving the amplifier. Red while on.
  - **AMP off** = normal QRP operation, up to the radio's full 5 W, using the QRP CAL power table.
  - **AMP on** = the radio uses the **AMP CAL** power table, set up once on the AMP CAL tab (4.6) to drive your amplifier correctly.

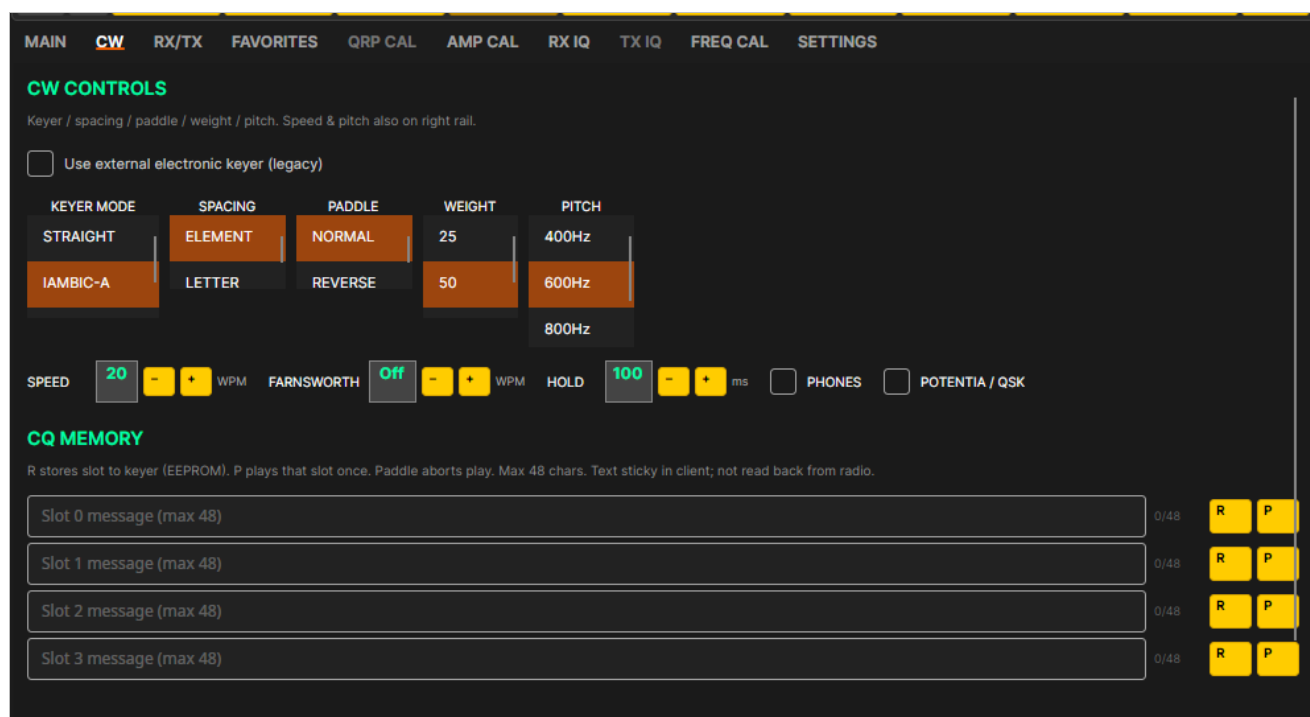
**▲ WARNING: Don't overdrive your amplifier.** Multus amplifiers are designed to give full output with about 75 % of the radio's power. Feeding the full 5 W QRP output into the amplifier overdrives it. Always turn **AMP on** when the amplifier is in line, and calibrate the AMP table first (4.6).

- 32. **CMP.** Speech compressor on/off.
- 33. **S/W.** Opens the Spectrum / Waterfall Controls window (4.11).
- 34. **POWER slider** (0–100 %). The label changes to match what you're doing: **TUNE** (TUN on), **CW**, **AM**, **FM**, or **SSB** (LSB, USB and DIG-U). These are the same settings as on the RX/TX tab. The number is not watts: it is a percentage of the QRP CAL level you set (4.5).
- 35. **COMP** (0–24). Compressor amount. Phones path only.
- 36. **AGC** (0–1000 ms). AGC fast release: how quickly the AGC recovers.
- 37. **Mode buttons: USB, LSB, CW, AM, DIG-U, FM.** The active mode is darker.
  - **DIG-U** is upper sideband for digital modes. It switches audio to the Digital path and gives the extra 1.4k/1.0k Hi filters.
  - **FM** is narrow FM. By default it transmits 100 kHz below the receive frequency (repeater split), shown on VFO B.
- 38. **Simplex (no offset).** FM only. Check it to transmit on the receive frequency.
- 39. **NB / NR / AN / MON.** Noise blanker, noise reduction, automatic notch, and monitor (hear your own transmit audio), each on/off.
- 40. **Pls** (10–510). Noise blanker pulse width.
- 41. **Thr** (1–100). Noise blanker threshold.
- 42. **NR** (0–100). Noise reduction strength.
- 43. **CW – / speed / +.** Keyer speed in WPM (5–60).

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44. **CW pitch** (for example 600Hz). Each click: 400 → 600 → 800 → 1000 Hz. The CW tone you hear.
45. **LOG**. Opens the Debug Log window (4.12).

### 4.2 CW Tab



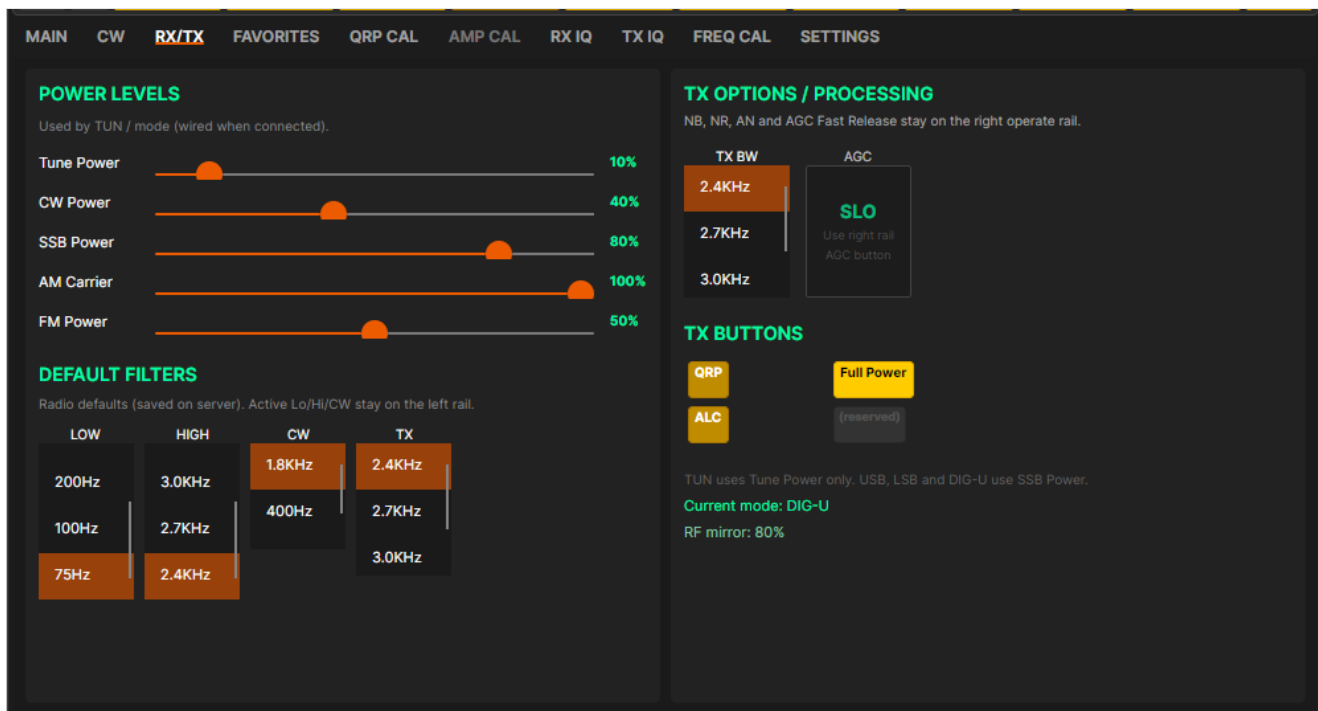
*CW tab*

1. **Use external electronic keyer (legacy)**. Check this if you key the radio with your own external keyer, or you have a Legacy (non-MKII) radio.
  - The radio's built-in keyer controls gray out. HOLD still works. An orange note reminds you of this.
  - The servers read this setting only when they start. After changing it, Disconnect and Connect again (with Launch), or use MSCC Stop and MSCC Start.
2. **KEYER MODE**. STRAIGHT, IAMBIC-A (default) or IAMBIC-B.
3. **SPACING**. ELEMENT or LETTER.
4. **PADDLE**. NORMAL or REVERSE (swaps dot and dash).
5. **WEIGHT**. 25, 50 (default) or 75.
6. **PITCH**. 400, 600, 800 or 1000 Hz. Same as the CW pitch button on the right rail.
7. **SPEED** - / +. 5–60 WPM. Same as the right rail.
8. **FARNSWORTH** - / +. Off, or 5–60 WPM. It adds extra space between letters and words **when playing CQ memories only**; your paddle sending isn't affected.

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9. **HOLD** – / +. 1–500 ms in steps of 10 (default 100). How long the radio stays in transmit after you stop keying.
10. **PHONES** and **POTENTIA** / **QSK**. Same as in MSCC for Windows. **Question for Ron:** what do PHONES and POTENTIA / QSK on the CW tab do? The Windows guide left these open too
11. **CQ MEMORY slots 0–3**. Each slot has a text box, a character count, **R** and **P**.
  - Type up to 48 characters per slot.
  - **R** stores the text in the radio's keyer.
  - **P** plays it once. Touch the paddle to stop playback.
  - R and P work only while connected. The text you type is remembered by MSCC UI; the radio can't send it back.
12. **Memory status line**. For example “Stored slot 0 (20 chars)” or “Not connected”.

### 4.3 RX/TX Tab



RX/TX tab

#### POWER LEVELS

1. **Tune Power** (0–100 %). Sets the drive for TUNE only (the TUN carrier). No other mode uses it.
2. **CW Power** (0–100 %).
3. **SSB Power** (0–100 %). LSB, USB and DIG-U.
4. **AM Carrier** (0–100 %).
5. **FM Power** (0–100 %).



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### DEFAULT FILTERS

6. **LOW, HIGH, CW, TX.** The radio's stored default filters (saved on the servers). The filters you use day to day are the Lo, Hi and CW buttons on the left rail.

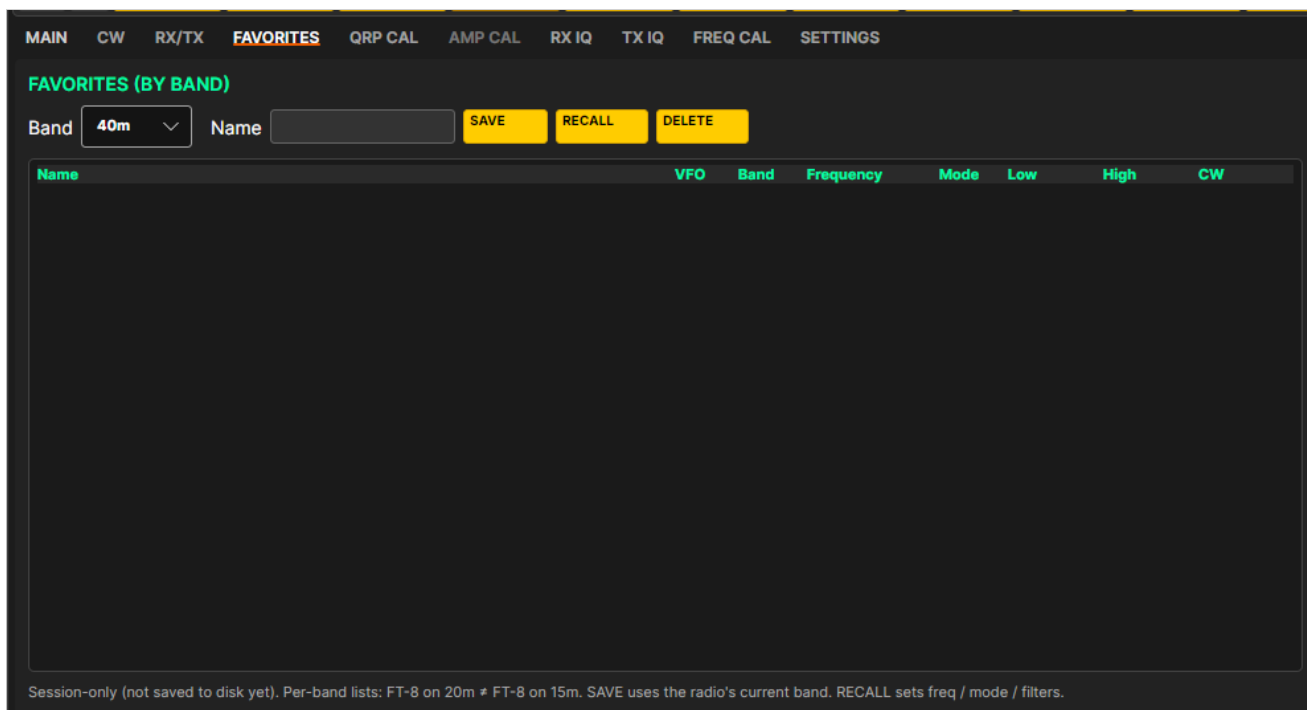
### TX OPTIONS / PROCESSING

7. **TX BW.** Transmit bandwidth: 2.4k, 2.7k, 3.0k or 5.5k (scroll the list for 5.5k). In MSCC UI this setting is only on this tab, not on the right rail.
8. **AGC.** Shows the current AGC speed. Change it with the AGC button on the right rail.

### TX BUTTONS

9. **QRP.** Barefoot power, the same as **AMP** off. 100 % gives the 5 W you set in QRP CAL.
10. **Full Power.** The amplifier setting, the same as **AMP** on. The radio uses the AMP CAL table for the band (4.6); it never drives above your QRP CAL level.
11. **ALC.** Transmit automatic level control on/off. Keep ALC on, especially for digital modes, and set the digital level so the ALC meter is just into the yellow. With ALC off, it is easy to overdrive and splatter.
12. **Notes** at the bottom: "TUN uses Tune Power only. USB, LSB and DIG-U use SSB Power.", the current mode, and the power setting in use for it ("RF mirror").

## 4.4 FAVORITES Tab

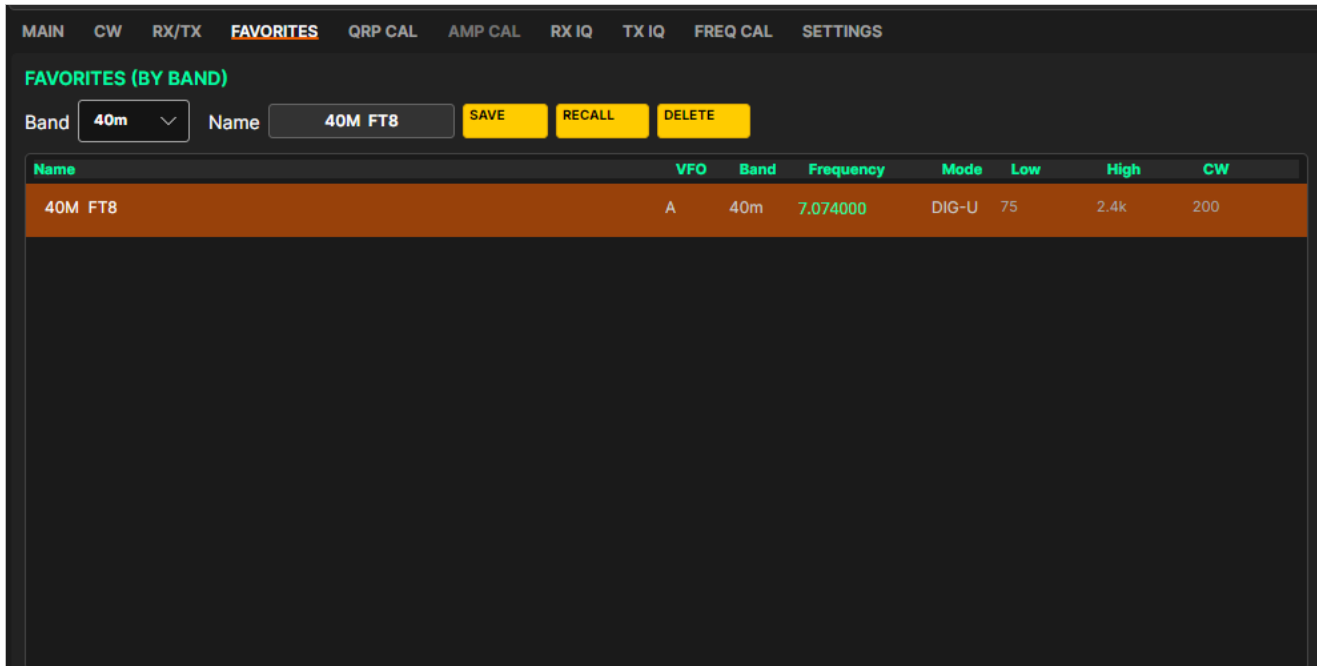


*FAVORITES tab, empty*

1. **Band.** Shows the favorites saved for one band. Each band has its own list, so “FT8” on 20 m and “FT8” on 15 m are separate favorites.
2. **Name.** Name for a new favorite (for example “40M FT8”).
3. **SAVE.** Saves the current frequency, mode, filters and VFO under this name, on the band the radio is on now.
4. **RECALL.** Returns to the selected favorite (frequency, mode and filters).
5. **DELETE.** Removes the selected favorite.
6. **Favorites list.** Name, VFO, band, frequency, mode and filters. Click a row to select it.



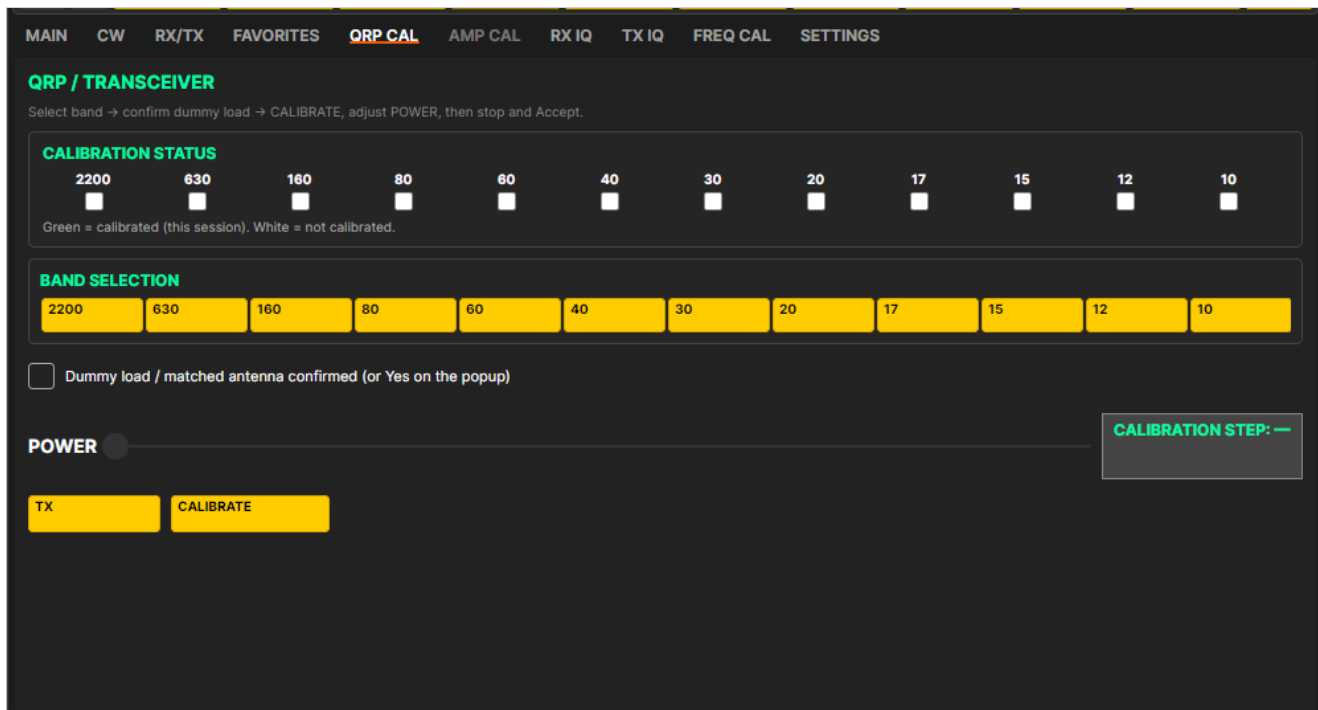
## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)



*FAVORITES tab with one saved favorite (40M FT8, 7.074000, DIG-U)*

Favorites are saved. They are still there after you close and reopen MSCC UI (they are kept in `~/.config/MSCC/mscc-favorites.ini`, Appendix D). The note at the bottom of the tab still says “Session-only (not saved to disk yet)”; that note is out of date and will be corrected in a later version.

### 4.5 QRP CAL Tab (power calibration; AMP must be off)



*QRP CAL tab*

This tab sets the radio's output power on each band.

**Do QRP CAL first, then AMP CAL.** On each band, calibrate QRP CAL (this tab, AMP off) before AMP CAL (4.6, AMP on). The AMP table is built on top of the QRP level, so doing them in the other order gives the wrong amplifier drive.

Your calibrated power is kept by the servers, on the computer the radio is plugged into, not in MSCC UI. That way the same calibration is also used when you operate remotely.

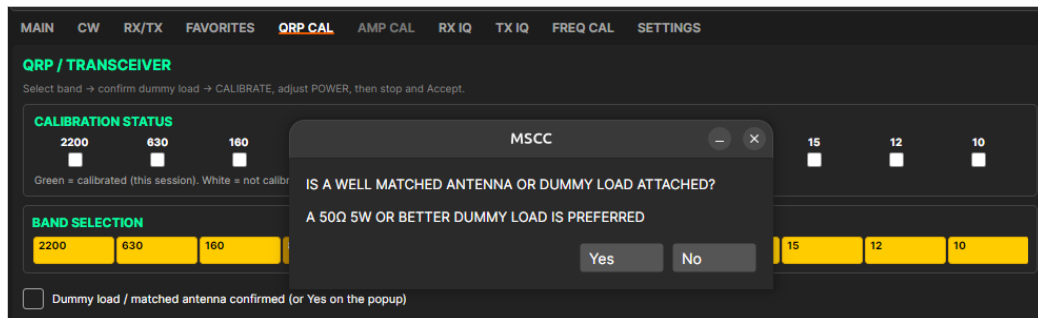
- You need a 50  $\Omega$  dummy load rated for 5 W or more, and a wattmeter.
  - **Do not set the power above 5 W. More than 5 W causes spurious emissions.** This applies to all current models.
1. **CALIBRATION STATUS lamps.** One per band. Green = calibrated **this session**, white = not calibrated yet this session. The lamps show only what you did since MSCC UI was opened, so they are all white at the start; your saved calibration is still in use.
  2. **BAND SELECTION.** Pick the band to calibrate; the radio tunes to that band's calibration frequency.
  3. **Dummy load / matched antenna confirmed.** Check this, or answer **Yes** when MSCC asks, to confirm a dummy load or matched antenna is connected.

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4. **POWER slider** (0–100). Adjusts the output. Works only while CALIBRATE is on.
5. **CALIBRATION STEP.** The current setting as reported by the radio.
6. **TX.** Sends a carrier at the calibration power; reads **TX ON** while on. Turn it off before CALIBRATE.
7. **CALIBRATE.** Starts calibration (reads **CALIBRATING**): the radio transmits and the POWER slider comes alive, starting from 0. Click it again to stop, then choose Accept, Reject or Cancel.

### Procedure:

1. Turn AMP off.
2. Connect the dummy load and wattmeter.
3. Open QRP CAL and pick a band.
4. Click **CALIBRATE**. The first time in a session, MSCC asks “**IS A WELL MATCHED ANTENNA OR DUMMY LOAD ATTACHED?**” Click **Yes** if it is. (No cancels.)



*Dummy load question, asked once per session*

5. Move **POWER** up until the wattmeter reads 5 W (not more).

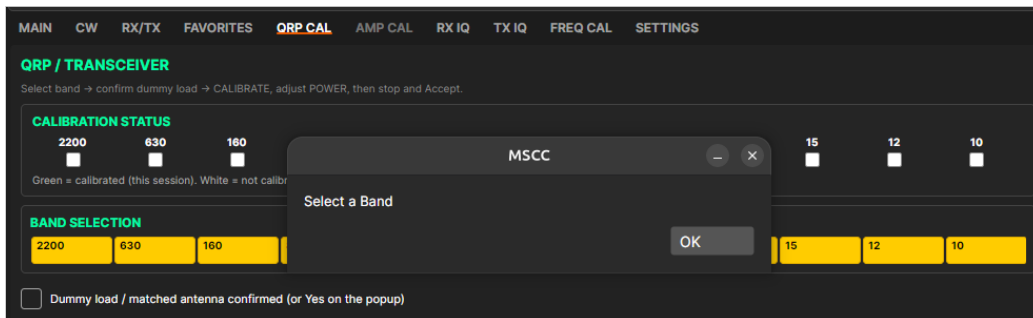
**The first slider move can take a few seconds.** The first time you move the POWER slider after starting CALIBRATE, it can take a few seconds before the power output rises. Nudge the slider up a little, wait until output begins, then set the power.

6. Click **CALIBRATE** again. A row appears on the tab: “**Accept this calibration?**”
  - **Accept** saves it and the lamp turns green.
  - **Reject** marks the band not calibrated.
  - **Cancel** puts back the previous setting.
7. Repeat for each band.

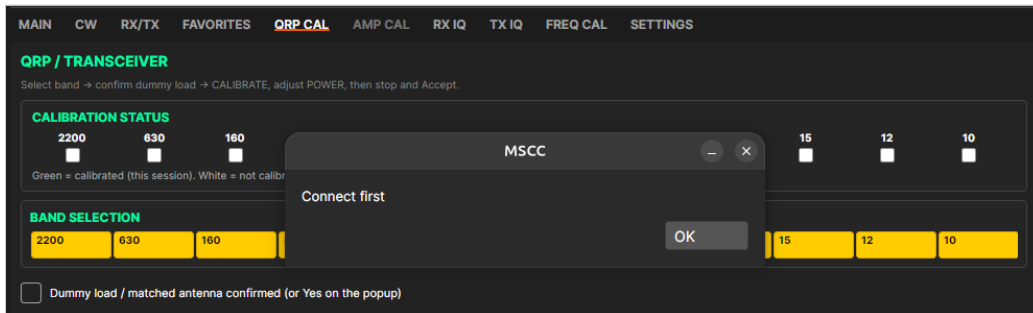
## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

Messages you may see here (see also Appendix A):

- **“Select a Band”**: pick a band first.
- **“Connect first”**: MISC UI isn’t connected to the radio. Press Connect on the left rail. (If you haven’t picked a band or answered the dummy load question yet, those come first.)
- **“TX ON. SET TX OFF BEFORE CALIBRATE”**: click TX to turn the test carrier off, then CALIBRATE.
- **“Finish or cancel CALIBRATE first.”**: you pressed TX while calibrating.

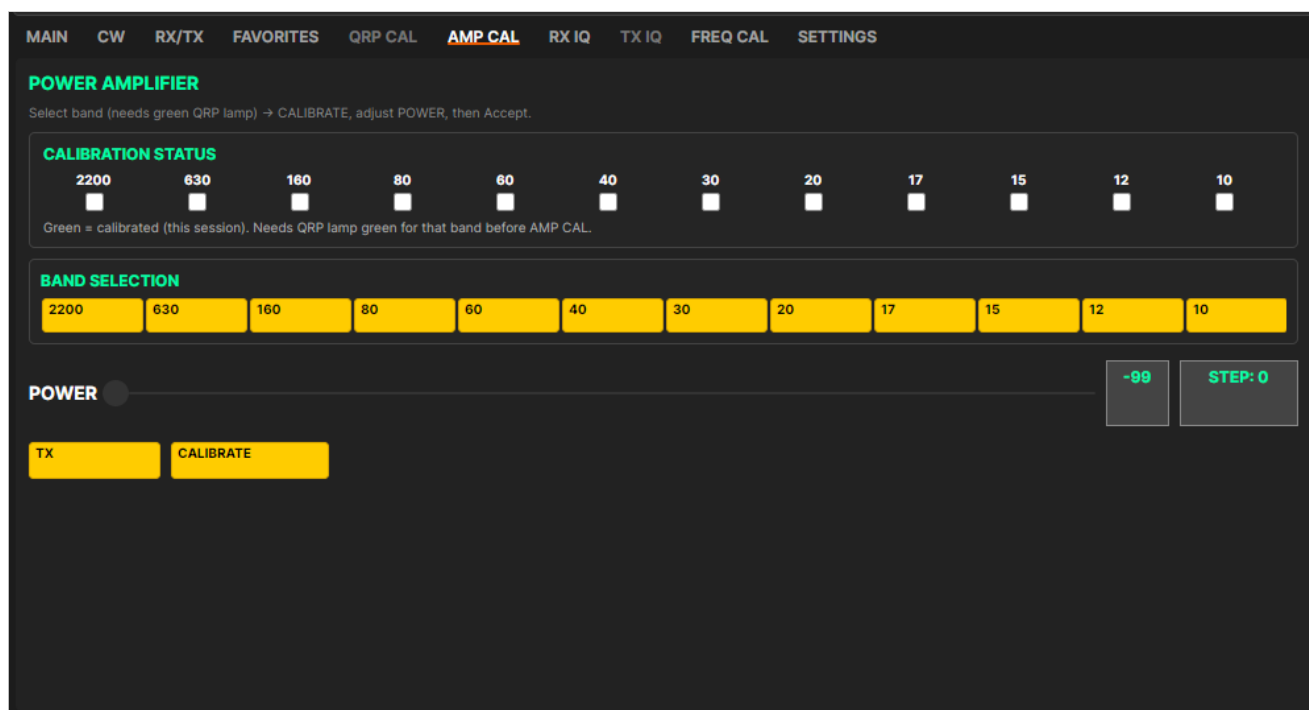


*QRP CAL: Select a Band*



*QRP CAL: Connect first*

## 4.6 AMP CAL Tab (AMP must be on)



*AMP CAL tab*

Use this tab to set up the **AMP power table**, the drive level the radio uses when **AMP** is on. Once each band is calibrated here, you can switch between QRP (AMP off, 5 W) and QRO (AMP on, amplifier) without overdriving the amplifier.

**Do QRP CAL first, then AMP CAL.** Calibrate a band on the QRP CAL tab (4.5) before you calibrate it here. Like QRP CAL, the AMP calibration is kept by the servers, so it is also used for remote operation.

**⚠ WARNING:** Multus amplifiers reach full output with about 75 % of the radio's power. Full 5 W QRP drive overdrives them. Calibrate AMP CAL for each band you use with the amplifier, and keep **AMP on** whenever the amplifier is in line.

1. **CALIBRATION STATUS lamps.** Green = calibrated this session. As on QRP CAL, the lamps show only the current session. A band's **QRP CAL** lamp must be green (QRP CAL done since MSCC UI was opened) before MSCC lets you AMP CAL it.
2. **BAND SELECTION.**
3. **POWER slider** (-99 to 0). Works only while CALIBRATE is on. It starts at -99 (lowest drive).
4. **Value readout** and **STEP readout.**
5. **TX.** Sends a tune carrier through the amplifier path.

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6. **CALIBRATE**. Starts and stops calibration; then “Accept this AMP calibration?” with **Accept**, **Reject** or **Cancel**, the same as QRP CAL.

### Procedure:

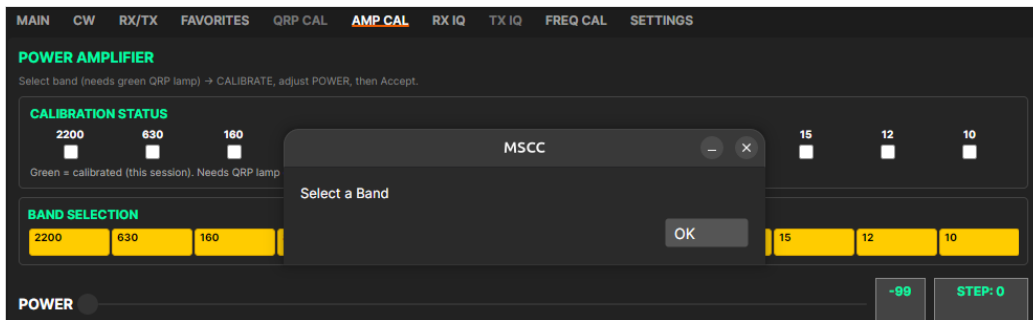
1. Do QRP CAL for the band first (4.5), so its QRP lamp is green. Always QRP CAL first, then AMP CAL.
2. Connect the amplifier to a dummy load, with a wattmeter on the amplifier output. Turn **AMP** on.
3. Open AMP CAL, pick the band and click **CALIBRATE**. Answer **Yes** to the dummy load question if asked.
4. Raise **POWER** slowly until the amplifier just reaches its rated output.

**The first slider move can take a few seconds.** The first time you move the POWER slider after starting CALIBRATE, it can take a few seconds before the power output rises. Nudge the slider a little, wait until output begins, then set the power.

5. Click **CALIBRATE** again and choose **Accept**.

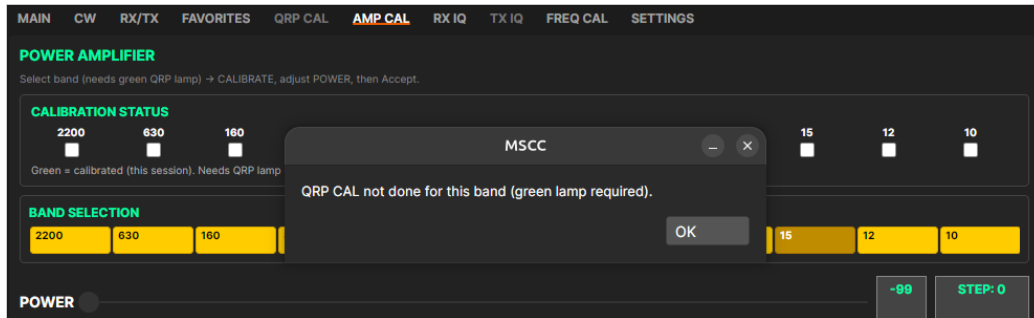
### Messages you may see here:

- “**Select a Band**”: pick a band first.
- “**QRP CAL not done for this band (green lamp required).**”: do QRP CAL for this band first, in the same session.
- “**Connect first**”: press Connect.



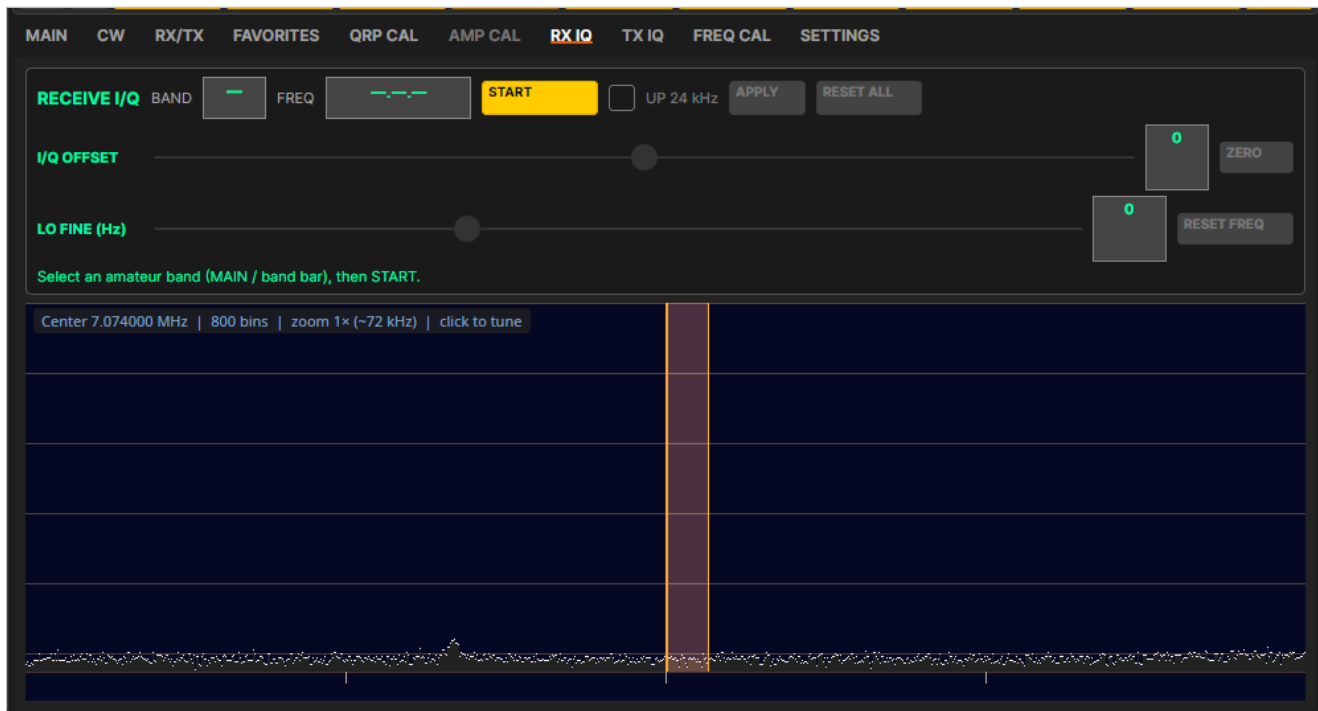
*AMP CAL: Select a Band*

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*AMP CAL: QRP CAL not done for this band*

### 4.7 RX IQ Tab (receive image balance)



*RX IQ tab*

First choose an amateur band on the MAIN tab. This doesn't work in general coverage (USER).

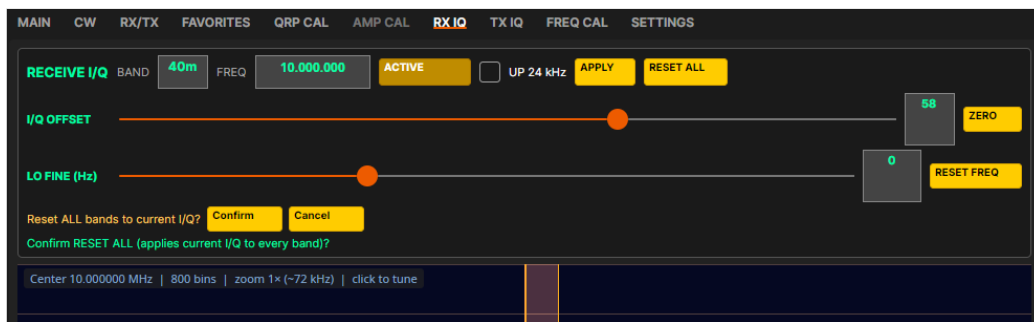
1. **BAND** and **FREQ**. The band and frequency being balanced.
2. **START** / **ACTIVE**. Starts or ends the session. The other controls work only while ACTIVE.
3. **UP 24 kHz**. Shifts the receiver up 24 kHz to help you find the image.
4. **APPLY**. Saves the value for this band.
5. **RESET ALL**. Copies the current value to **all** bands, after you confirm.
6. **I/Q OFFSET** (–200 to +200). Adjust for the smallest image on the spectrum.
7. **ZERO**. Sets the offset to 0 and saves it.
8. **LO FINE (Hz)** (–1000 to +2009). Fine retune during the session.

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9. **RESET FREQ.** Clears LO FINE.
10. **Status line.** Tells you what to do next.
11. **Spectrum.** Live view to watch the image.

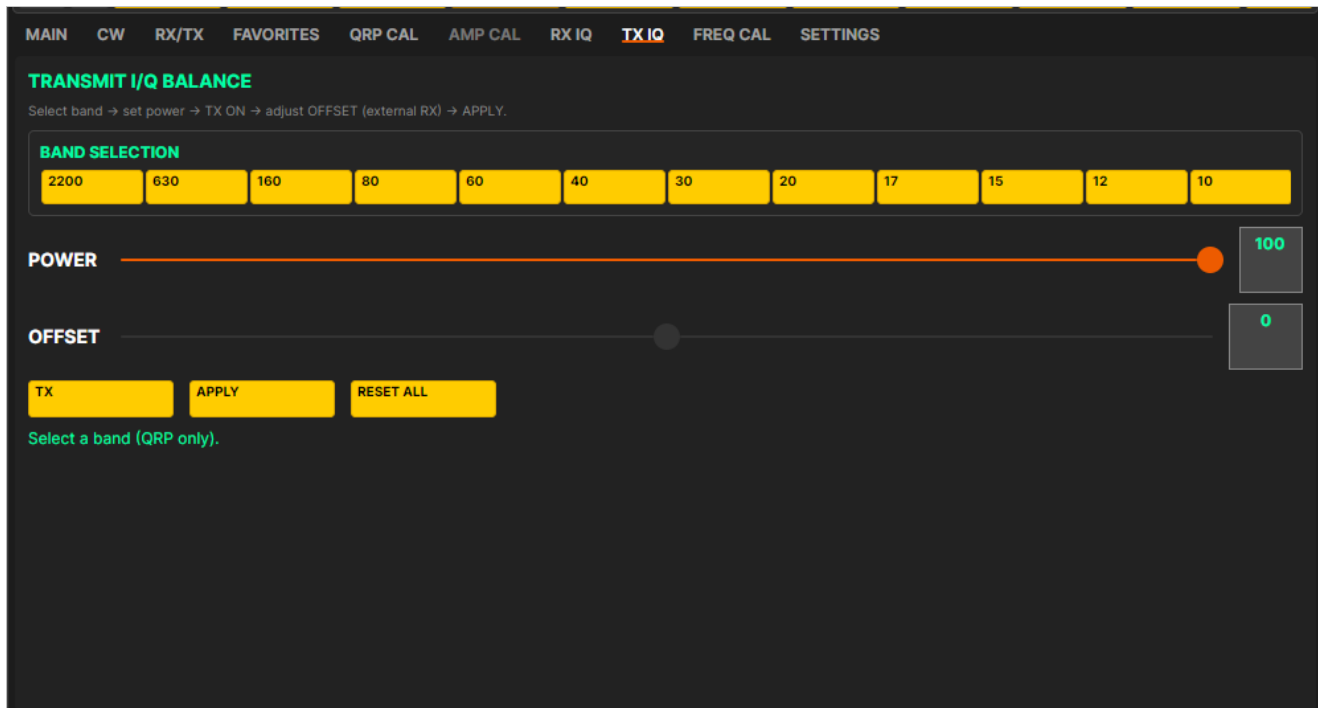
**Procedure:** press **START**, find the image signal on the spectrum (UP 24 kHz helps), move **I/Q OFFSET** until the image drops into the noise, then press **APPLY**. The best value is different for every radio and band.

**RESET ALL** asks on the tab itself: “Reset ALL bands to current I/Q?” Click **Confirm** to copy the current value to every band, or **Cancel**.



*RX IQ: RESET ALL asks Confirm or Cancel*

### 4.8 TX IQ Tab (transmit image balance; AMP must be off)



*TX IQ tab*

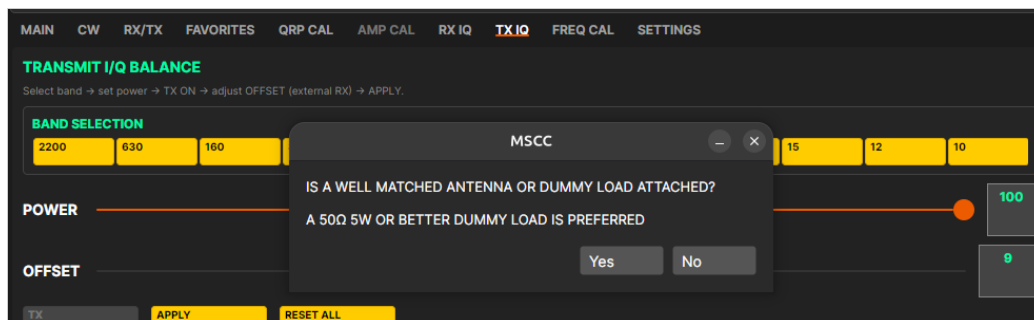
## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)

You need a second receiver to hear or see the transmit image. Use a dummy load, and keep the second receiver's antenna well away so it doesn't overload. Tune the second receiver about 24 kHz from the displayed frequency to find the image.

1. **Instructions.** Select band → set power → TX ON → adjust OFFSET (external RX) → APPLY.
2. **BAND SELECTION.** Tunes to that band's TX IQ frequency. Locked while TX is on.
3. **POWER** (0–100, default 100).
4. **OFFSET** (–200 to +200). Adjust for the weakest image. Works only while TX is on.
5. **TX / TX ON.** Starts or stops the test carrier.
6. **APPLY.** Saves the value for this band, after asking.
7. **RESET ALL.** Returns **all** bands to the factory values, after you confirm. TX must be off.
8. **Status line.**

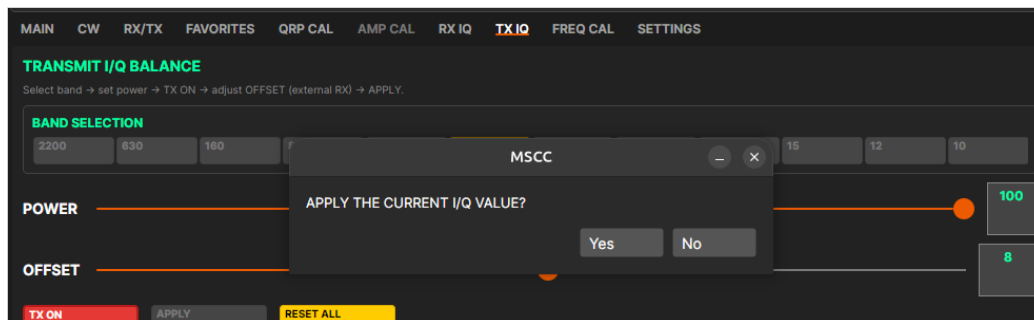
### Procedure:

1. Turn AMP off, select a band and press **TX**. Answer **Yes** to the dummy load question if asked.



*TX IQ: dummy load question*

2. Watch the second receiver and move **OFFSET** until the image is weakest.
3. Press **APPLY**. MSCC asks “**APPLY THE CURRENT I/Q VALUE?**” Choose **Yes** to save the value for this band or **No** to keep the old one.

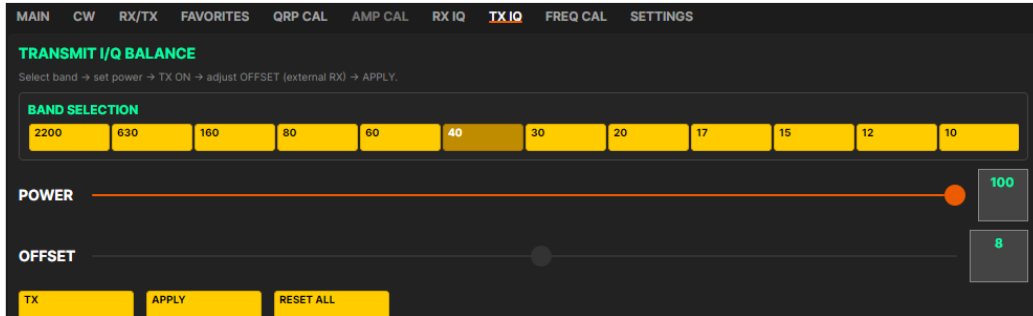


*TX IQ: APPLY THE CURRENT I/Q VALUE?*

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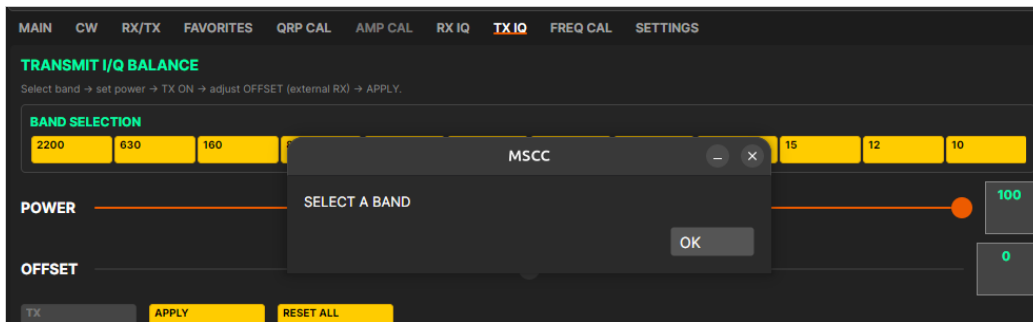
4. Press TX again to turn the carrier off.

**RESET ALL** (only if needed) asks on the tab: “**Factory-reset ALL I/Q bands?**” Click **Confirm** only if you want to throw away the saved values for **every** band; otherwise **Cancel**.



*TX IQ: RESET ALL asks Confirm or Cancel*

If you press TX without picking a band you’ll see “**SELECT A BAND**”; with AMP on you’ll see “**TX IQ balance requires QRP mode (AMP off).**”



*TX IQ: SELECT A BAND*

### 4.9 FREQ CAL Tab (frequency calibration)

**While a calibration is running, don’t change tabs or settings until it finishes.** The tab shows a reminder during the run. MSCC won’t let you open QRP CAL, AMP CAL or TX IQ until the run is done.

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*FREQ CAL tab*

**Mode on this tab.** Opening FREQ CAL switches the radio to **CW** with a **600 Hz pitch** and the **200 Hz** CW filter, which is what the measurement needs. When you leave the tab, MSCC puts back your mode, filter and pitch. While MANUAL is on, you can't leave the tab; finish MANUAL first.

1. **LOOSE / TIGHT.** How close the measurement must get before it calls the job done. LOOSE locks more easily on a weak carrier; TIGHT is stricter. AUTO and CHECK both use this setting.
2. **AUTO.** Automatic calibration against a steady carrier on your current frequency. A row appears on the tab, **AUTO: Coarse ±250 Hz / Fine ±50 Hz / Cancel**, and the status reads "Choose COARSE or FINE...".
3. **MANUAL.** Manual adjustment with the **MANUAL steps** - / + buttons (up to ±100 steps). Click MANUAL again to finish; MSCC asks "**Accept manual calibration?**" (**Accept** keeps it, **Reject** throws it away).
4. **CHECK.** Measures the error without changing anything. CHECK only looks within ±50 Hz. If the radio is further off than that (for example right after RESET), CHECK reports **CHECK FAILED — Error may be more than 50 Hz. Run AUTO (COARSE).**
5. **RESET.** Returns to the factory calibration, after you confirm on the tab ("**Reset frequency calibration?**", Confirm or Cancel).
6. **STOP.** Stops an AUTO or CHECK run. The calibration is not changed.
7. **ST: status.** RUNNING COARSE / RUNNING FINE / CHECKING, then AUTO COMPLETED or CHECK COMPLETED with a number in Hz, or FAILED. After CHECK, the number is the

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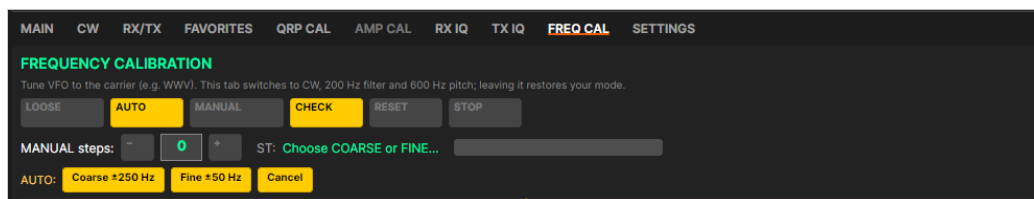
radio's error right now. After AUTO, it's the error AUTO found **before** it corrected it; run CHECK to see where you are now.

8. **Progress bar.** Fills as the run goes.

9. **Spectrum.** Live view.

### Procedure:

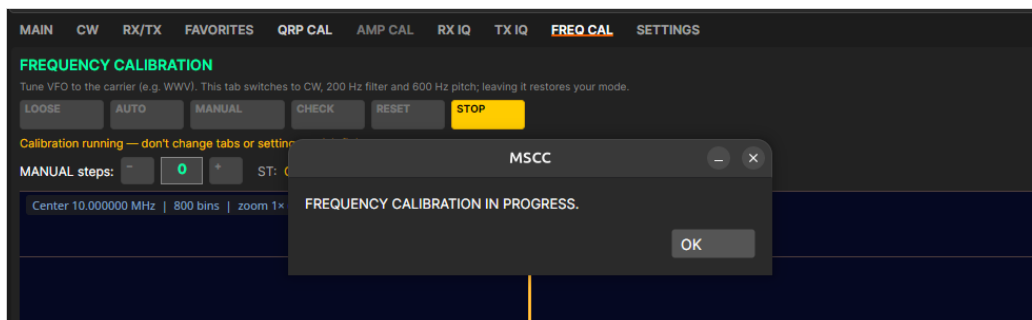
1. Click **USER** on the band bar until you reach a strong time station, S7 or better (WWV or CHU on HF).
2. Open **FREQ CAL**. The tab switches to CW for you.
3. Click **AUTO**, then **Coarse ±250 Hz**. Wait for **AUTO COMPLETED**.



*AUTO: choose Coarse ±250 Hz, Fine ±50 Hz, or Cancel*

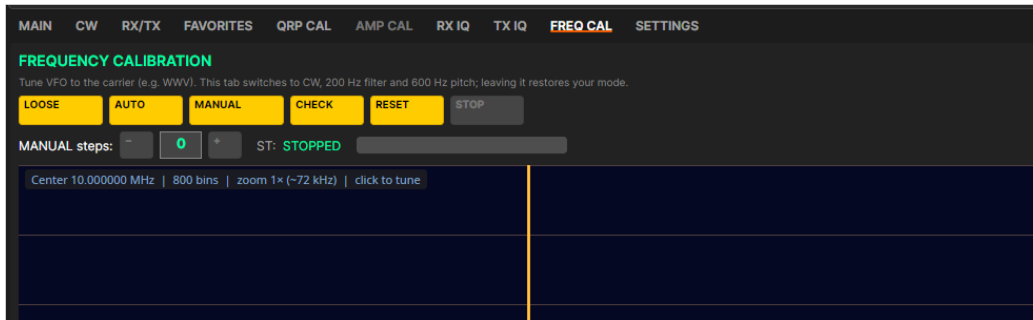
4. Click **AUTO** again, then **Fine ±50 Hz**. Wait for **AUTO COMPLETED**.
5. Click **CHECK** to confirm. An error of a few Hz (for example -1 Hz) is a good result.
6. Stay on this tab until each run finishes.

If you try to open QRP CAL, AMP CAL or TX IQ during a run, MSCC says “**FREQUENCY CALIBRATION IN PROGRESS.**” Click OK and wait, or press **STOP**.

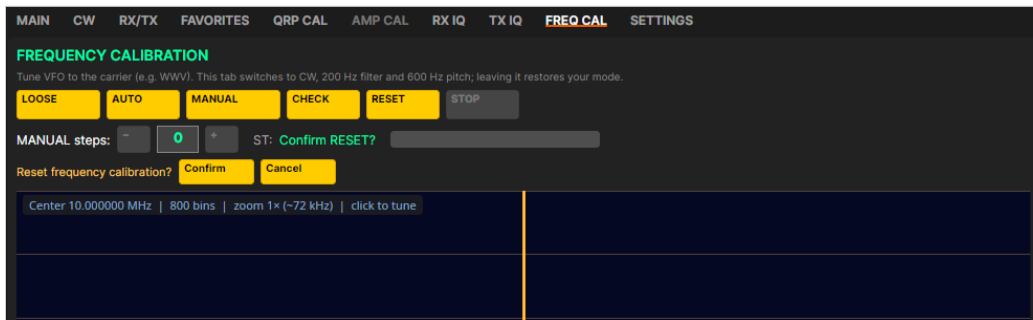


*FREQUENCY CALIBRATION IN PROGRESS.*

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*After STOP: ST: STOPPED*



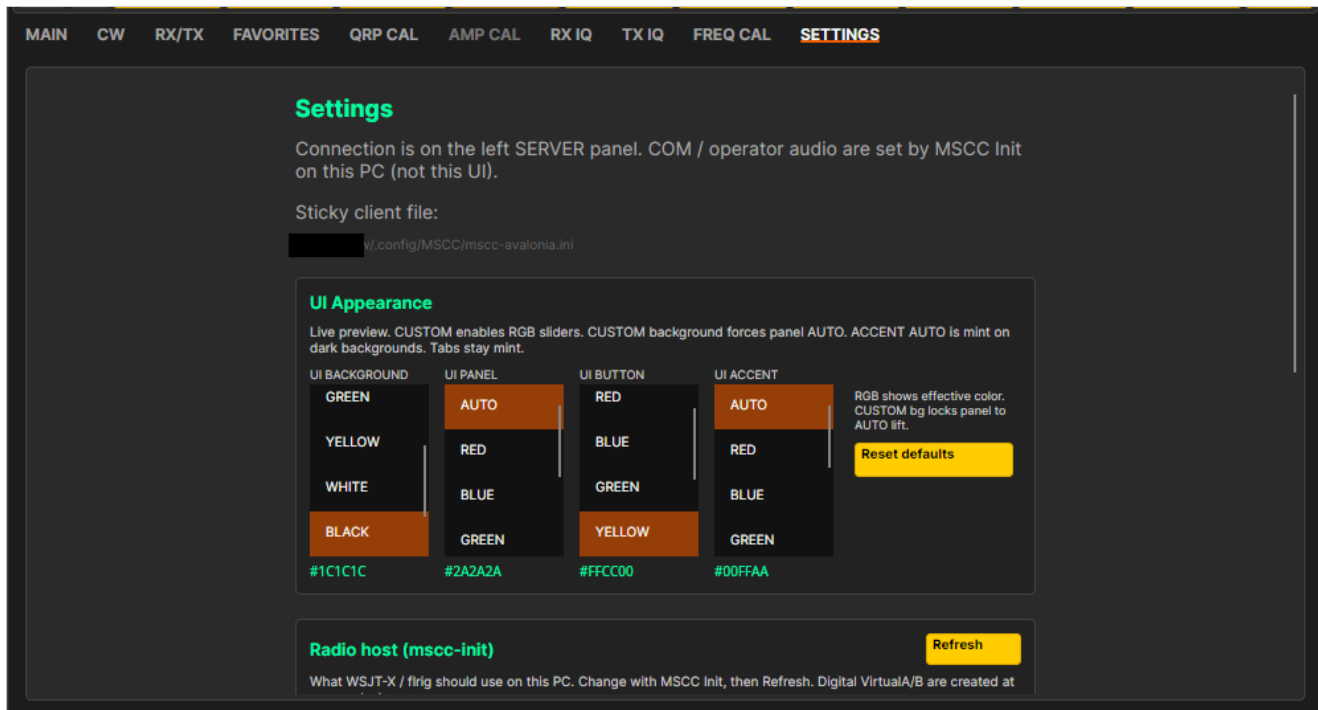
*RESET asks Confirm or Cancel on the tab*

After a RESET, pressing CHECK first asks: “Calibration was reset. Run AUTO first. Click OK to run CHECK anyway, or Cancel.”

### 4.10 SETTINGS Tab

The SETTINGS page scrolls, so it is shown in three pictures. Unlike MSCC for Windows, the serial (COM) port and audio devices are **not** set here: they are set with **MSCC Init** (2.4), and SETTINGS only shows them.

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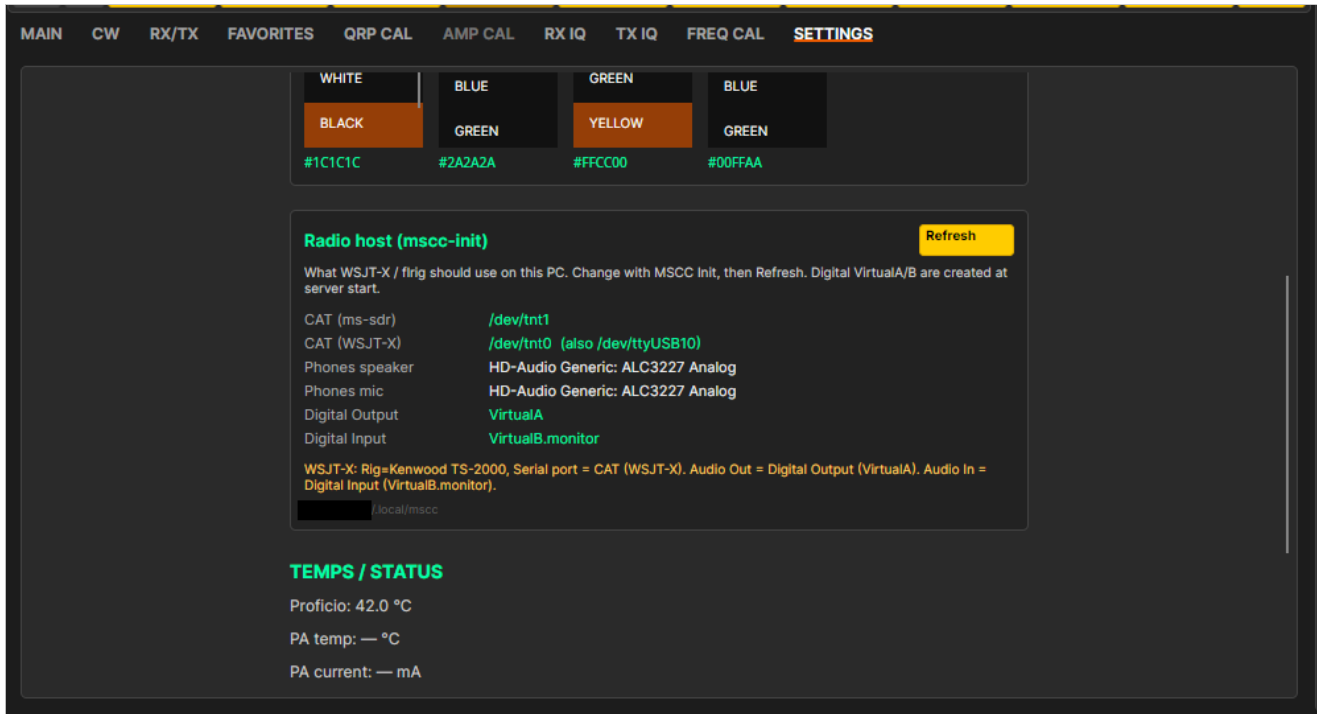
### *SETTINGS, part 1: settings file and UI Appearance*

1. **Sticky client file.** Where MSCC UI keeps its settings (~/.config/MSCC/mscc-avalonia.ini).

### **UI Appearance** (changes show immediately)

2. **UI BACKGROUND.** Window background color, or CUSTOM (R/G/B sliders appear).
3. **UI PANEL.** Panel color; AUTO is a little lighter than the background.
4. **UI BUTTON.** Button color (YELLOW is the classic Multus gold). CUSTOM gives R/G/B sliders.
5. **UI ACCENT.** Color of headings and readouts. CUSTOM gives R/G/B sliders.
6. **Color codes.** The color actually in use, under each list.
7. **Reset defaults.** Background BLACK, panel AUTO, button YELLOW, accent AUTO.

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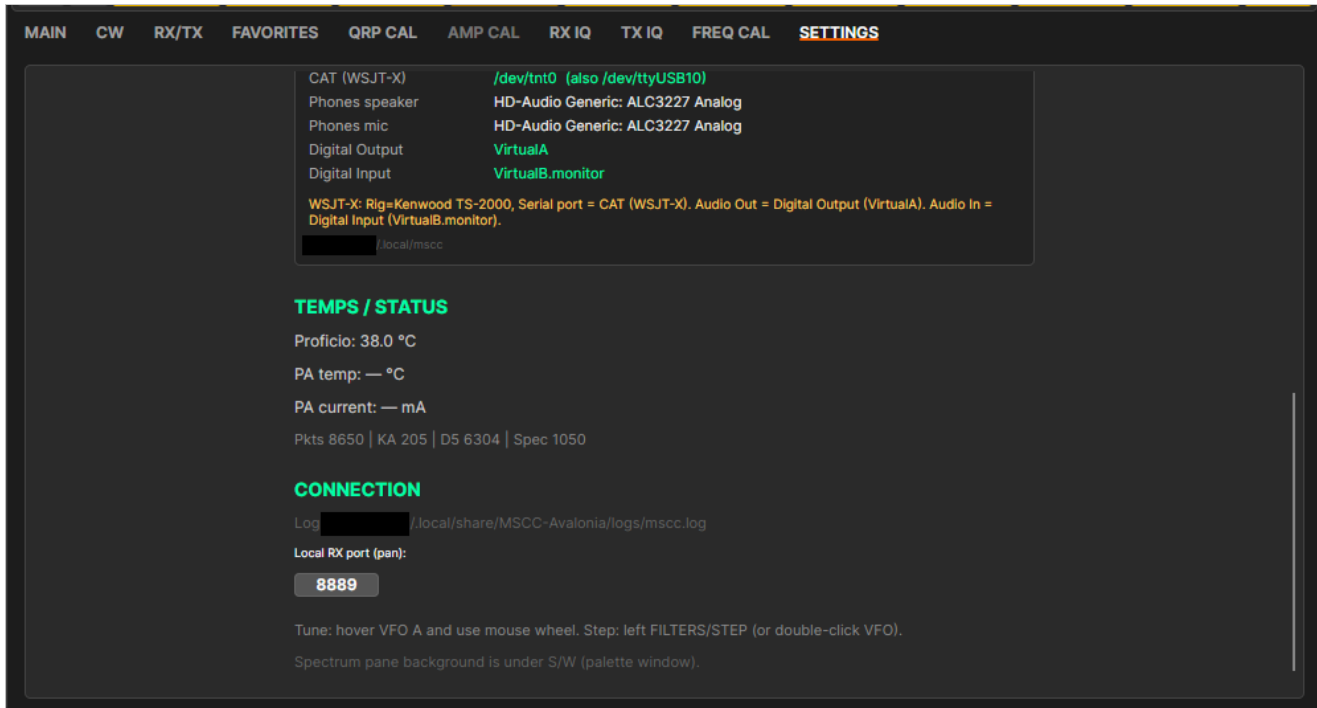


SETTINGS, part 2: Radio host (mscc-init)

**Radio host (mscc-init)** (read-only; change these with MSCC Init, then press **Refresh**)

8. **CAT (ms-sdr)**. The serial port the servers use.
9. **CAT (WSJT-X)**. The port your digital program should use (the other end of the pair). In the example: /dev/tnt0 (also /dev/ttyUSB10).
10. **Phones speaker** and **Phones mic**. Your operator audio devices.
11. **Digital Output** (VirtualA) and **Digital Input** (VirtualB.monitor). The digital audio devices for WSJT-X.
12. **WSJT-X hint** (orange). The settings to use in WSJT-X (Appendix C).
13. **Refresh**. Reads the MSCC Init settings again.

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*SETTINGS, part 3: temperatures and connection*

### TEMPS / STATUS

14. **Transceiver temperature**, labeled with the radio name (for example “Proficio: 38.0 °C”).
15. **PA temp / PA current**. Reserved for later.
16. **Counters** (Pkts, KA, DS, Spec). Data arriving from the servers.

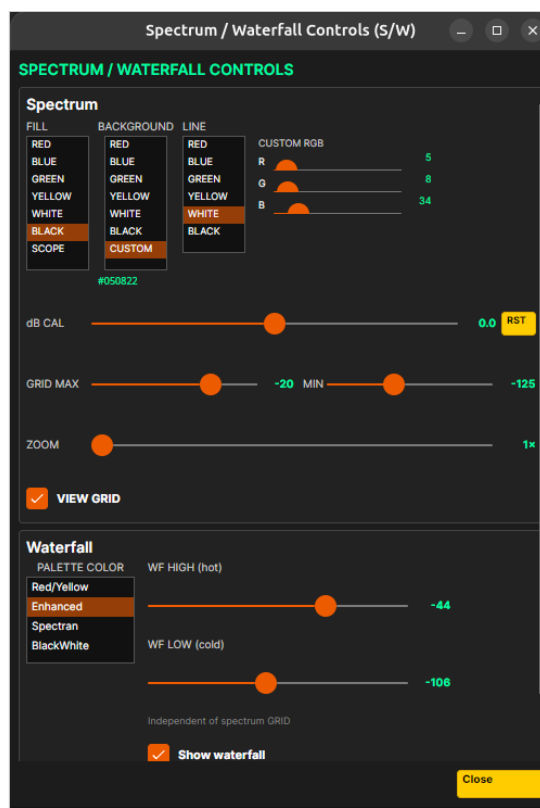
### CONNECTION

17. **Log** file location (~/.local/share/MSCC-Avalonia/logs/mscc.log).
18. **Local RX port (pan)**. The port MSCC UI listens on for the spectrum (default 8889). Leave it alone unless told otherwise.

## 4.11 Spectrum / Waterfall Controls window (S/W)

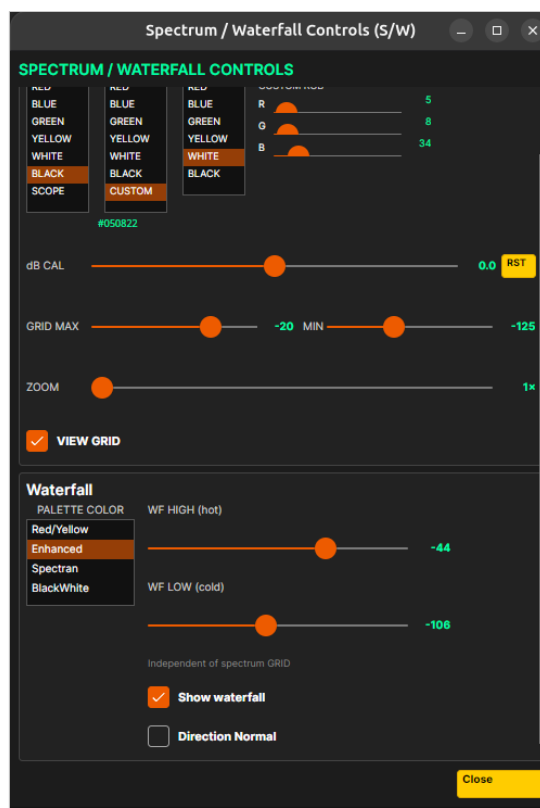
Open it with S/W on the right rail. It is shown in two pictures because the window scrolls.

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*S/W window, top*

## MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)



*S/W window, scrolled to the bottom*

### Spectrum

1. **FILL.** Color under the spectrum trace (SCOPE = line only, the classic look).
2. **BACKGROUND.** Spectrum background; CUSTOM shows the **CUSTOM RGB** sliders (R, G, B).
3. **LINE.** Trace color.
4. **dB CAL** ( $\pm 20$  dB) and **RST**. Trims the dB scale; RST sets it back to 0.
5. **GRID MAX** and **MIN**. Top and bottom of the spectrum display in dBm.
6. **ZOOM** ( $1\times$ – $4\times$ ). Same as ZOOM on the left rail.
7. **VIEW GRID.** Shows the horizontal dB grid lines.

### Waterfall

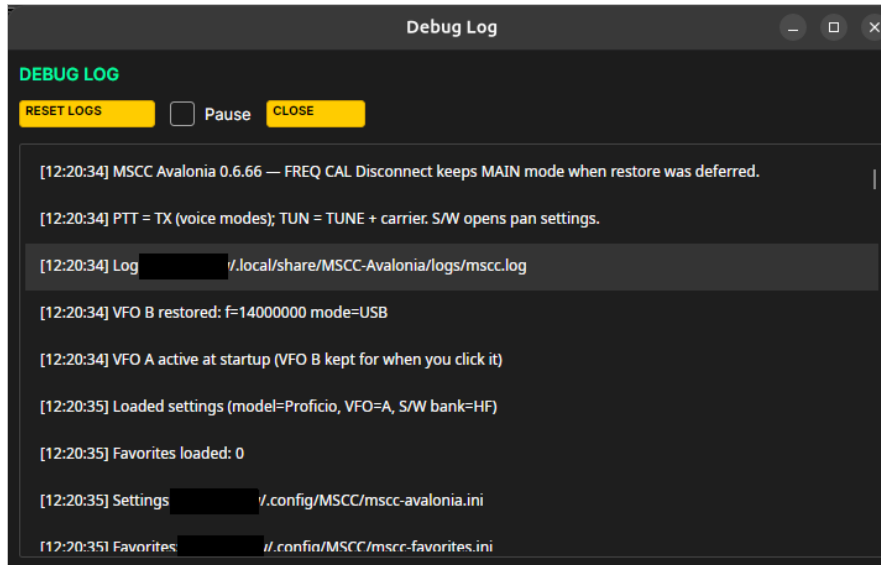
8. **PALETTE COLOR.** Red/Yellow, Enhanced, Spectran or BlackWhite. Kept separately for HF (Proficio, Ultimus) and LF (Geminus).
9. **WF HIGH (hot).** Signals this strong show in the hottest color.
10. **WF LOW (cold).** Background level of the waterfall. These two are independent of the spectrum grid.
11. **Show waterfall.** Turns the waterfall on or off.

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12. **Direction Normal.** Checked = oldest line at the top, newest scrolls down. Unchecked = reversed.
13. **Close.**

### 4.12 Debug Log window

Open it with **LOG** on the right rail.



*Debug Log window*

1. **RESET LOGS.** Clears the display and starts fresh.
2. **Pause.** Freezes the display; the log file keeps recording.
3. **CLOSE.** Closes the window.
4. **Log list.** Time-stamped messages: what MSCC UI did and what the servers reported.

The log file is `~/.local/share/MSCC-Avalonia/logs/mscc.log` (the `~` means your home folder). The servers keep their own logs in `~/.local/mscc/`. Send both with any problem report, together with a picture of MSCC Status (3.4).

### 4.13 Keyboard and mouse tips

- MSCC UI has no menus and no keyboard shortcuts. A mouse with a wheel is strongly recommended, especially on laptops.
- Wheel over a VFO digit tunes that digit. Wheel over the VFO box (not a digit) tunes by Step. Double-click VFO A to change the Step. Wheel over a slider nudges it.
- Left-click the spectrum to tune.
- Hover over a control to see a short tip about it.

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### □ Appendix A: Pop-up Messages

These are the messages MSCC UI, MSCC Init, the MSCC menu and Firmware Upload can show during local operation, what they mean, and what to do. Most MSCC UI messages have a title of **MSCC** and an **OK** button.

#### MSCC UI: connecting

| Message                                                                                           | What it means                                                                     | What to do                                                                                                    |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| MSCC is not set up on this computer yet. Run MSCC Init (menu: MSCC Init), then try again.         | Connect with Launch found that MSCC Init hasn't been run (its files are missing). | It's a warning: click <b>OK</b> and MSCC UI carries on. Run MSCC Init (2.4) when you can.                     |
| Desktop notification: Servers started (recv, trans, ms-sdr).                                      | The servers started (from Launch or MSCC Start).                                  | Nothing; it fades by itself.                                                                                  |
| Desktop notification: Servers stopped.                                                            | The servers stopped (Disconnect with Launch, closing MSCC UI, or MSCC Stop).      | Nothing.                                                                                                      |
| Desktop notification: Start failed. Try Terminal: msc start / Stop failed. Try Terminal: msc stop | The start or stop command failed.                                                 | Run <code>msc start</code> (or <code>msc stop</code> ) in a Terminal to see why, and check MSCC Status (3.4). |

#### MSCC UI: calibration tabs

| Message                                                                                                 | Where                                                  | What it means / what to do                                                                                                     |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Select a Band / SELECT A BAND                                                                           | QRP CAL, AMP CAL, TX IQ                                | You pressed TX or CALIBRATE before picking a band. Click a band button first.                                                  |
| Connect first                                                                                           | QRP CAL, AMP CAL, TX IQ                                | MSCC UI isn't connected. Click OK, then press <b>Connect</b> on the left rail.                                                 |
| IS A WELL MATCHED ANTENNA OR DUMMY LOAD ATTACHED? A 50Ω 5W OR BETTER DUMMY LOAD IS PREFERRED (Yes / No) | QRP CAL, AMP CAL, TX IQ, first transmit of the session | <b>Yes</b> if a dummy load or matched antenna is connected; MSCC won't ask again this session. <b>No</b> cancels the transmit. |
| TX ON. SET TX OFF BEFORE CALIBRATE                                                                      | QRP CAL                                                | Click TX to turn the test carrier off, then CALIBRATE.                                                                         |

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| Message                                                                                        | Where                                                       | What it means / what to do                                                                                                                                                                |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finish or cancel CALIBRATE first.                                                              | QRP CAL, AMP CAL                                            | Finish the calibration (click CALIBRATE, then Accept / Reject / Cancel) before using TX.                                                                                                  |
| Turn TX OFF before changing band.                                                              | QRP CAL, AMP CAL                                            | Turn TX off, then pick the new band.                                                                                                                                                      |
| QRP CAL not done for this band (green lamp required).                                          | AMP CAL                                                     | Do QRP CAL for this band first (4.5).                                                                                                                                                     |
| TX IQ balance requires QRP mode (AMP off).                                                     | TX IQ                                                       | Turn AMP off.                                                                                                                                                                             |
| APPLY THE CURRENT I/Q VALUE? (Yes / No)                                                        | TX IQ                                                       | <b>Yes</b> saves the value for this band; <b>No</b> keeps the old one.                                                                                                                    |
| On-tab row: Accept this calibration? / Accept this AMP calibration? (Accept / Reject / Cancel) | QRP CAL, AMP CAL                                            | <b>Accept</b> saves and turns the lamp green; <b>Reject</b> marks the band not calibrated; <b>Cancel</b> puts back the previous setting (QRP CAL) or leaves the lamp unchanged (AMP CAL). |
| On-tab row: Reset ALL bands to current I/Q? (Confirm / Cancel)                                 | RX IQ                                                       | <b>Confirm</b> copies the current value to every band.                                                                                                                                    |
| On-tab row: Factory-reset ALL I/Q bands? (Confirm / Cancel)                                    | TX IQ                                                       | <b>Confirm</b> returns every band to factory values.                                                                                                                                      |
| On-tab row: AUTO: Coarse $\pm 250$ Hz / Fine $\pm 50$ Hz / Cancel                              | FREQ CAL                                                    | Choose Coarse first, then Fine (4.9).                                                                                                                                                     |
| On-tab row: Accept manual calibration? (Accept / Reject)                                       | FREQ CAL                                                    | <b>Accept</b> keeps the MANUAL setting; <b>Reject</b> throws it away.                                                                                                                     |
| On-tab row: Reset frequency calibration? (Confirm / Cancel)                                    | FREQ CAL                                                    | <b>Confirm</b> returns to the factory calibration. Then run AUTO (Coarse, then Fine).                                                                                                     |
| Calibration was reset. Run AUTO first. Click OK to run CHECK anyway, or Cancel.                | FREQ CAL                                                    | After a RESET, CHECK will probably fail. Click Cancel and run AUTO first.                                                                                                                 |
| FREQUENCY CALIBRATION IN PROGRESS.                                                             | Changing to QRP CAL, AMP CAL or TX IQ during a FREQ CAL run | Click OK and wait for the run to finish, or press STOP.                                                                                                                                   |
| EXIT MANUAL CALIBRATION FIRST.                                                                 | Leaving FREQ CAL while MANUAL is on                         | Click MANUAL again and Accept or Reject.                                                                                                                                                  |

### MSCC Init



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| Message                                                                                                 | What it means / what to do                                                                                                                |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| MSCC servers are running on this Pi ... Stop the servers and continue with MSCC Init? (Yes / No)        | Settings can't change while the servers use the audio and CAT port. <b>Yes</b> stops them and continues; <b>No</b> exits without changes. |
| Servers stopped. Continuing with configuration.                                                         | After Yes. Click OK.                                                                                                                      |
| Initialization cancelled. Stop servers first (menu: MSCC Stop, or: msc stop), then run MSCC Init again. | After No. Nothing was changed.                                                                                                            |
| Start MSCC servers now? (Yes / No)                                                                      | After Write config. <b>Yes</b> starts the servers; <b>No</b> finishes without starting them (start later with MSCC Start or Launch).      |
| MSCC servers started. Connect with the MSCC client on your PC.                                          | The servers are running. Open MSCC UI and press Connect.                                                                                  |

### MSCC Firmware Upload

| Message                                                         | What it means / what to do                                      |
|-----------------------------------------------------------------|-----------------------------------------------------------------|
| Bootload successful !! Power off, remove BOOT jumper, power on. | The update worked. Click OK, then follow Appendix B steps 9–11. |

### □ Appendix B: Updating the Radio Firmware (PSoC)

The radio's main processor (a Cypress PSoC) holds two programs:

- the **application**, which is the radio firmware you update;
- a small **bootloader** ("LOADER") that you normally never touch.

Updating the application with a .cyacd file never changes the bootloader.

**All radios must be updated to the latest firmware to use MSCC.** Update the firmware before you run a new version of MSCC for the first time. After you connect, MSCC UI's title bar shows the firmware version the radio has now (FW).

**Which file:** one per radio model, named for the radio and date, for example Proficio-MKII-PTT-20260822.cyacd. The Windows guide lists the current firmware version for each radio.

**Where to get the file:** the Linux and Raspberry Pi kits don't include the firmware files. Download the .cyacd file for your radio from the MSCC GitHub repository, <https://github.com/w4mmp-ron/mscc-station>, or from the group's Proton Drive folder (group members have the link). Save it somewhere easy to find, such as your Downloads folder.

**▲ Don't interrupt a firmware update.** Once you click **Program**, don't unplug the USB cable, turn off the radio or close the program until the success message appears. It takes only a few seconds.

#### B.1 Updating the firmware over USB

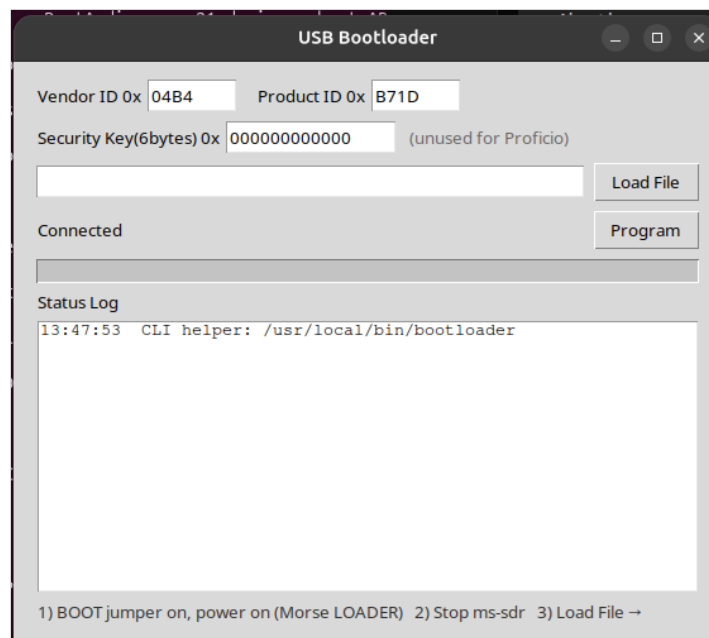
You need the radio, its USB cable plugged into the computer, and the .cyacd file for your radio. No extra driver is needed on Linux.

1. **Stop MSCC and the servers.** In MSCC UI press Disconnect (or close it). If the servers are still running, use **MSCC Stop** (or type `mscc stop`).
2. **Power the radio off.** Always move the shunt (jumper) with the radio powered **off**. Leave the USB cable connected.
3. **Put the shunt on the boot pins.** Find the 3-pin **BOOT1** header (labeled on the board's silkscreen). A small "U" printed on the silkscreen joins two of its pins: those are the boot pins. With the radio still off, put the small plastic shunt (jumper) across the two pins joined by the printed U. The header's location differs by radio; the Windows guide (Appendix A) has pictures of a Geminus MKII board.

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**Don't worry about the other pins.** The other pins are just grounds. The radio doesn't need a shunt to operate, and a shunt on the wrong pins doesn't hurt anything.

4. **Power the radio on.** The LED should blink **LOADER** in Morse code (very old firmware blinked "BOOT"). The radio is now ready to accept firmware. **If the LED isn't blinking**, the shunt is on the wrong pins: power off, move it to the two pins joined by the U, and try again.
5. **Start MSCC Firmware Upload** from the MSCC menu (or type `bootloader-gui` in a Terminal). Its window is titled **USB Bootloader**. The steps are also printed along the bottom of the window.
6. **Wait for Connected.** The line above the progress bar must show **Connected** before you go on. If it doesn't, check that the USB cable is plugged in and the LED is blinking **LOADER** (step 4).

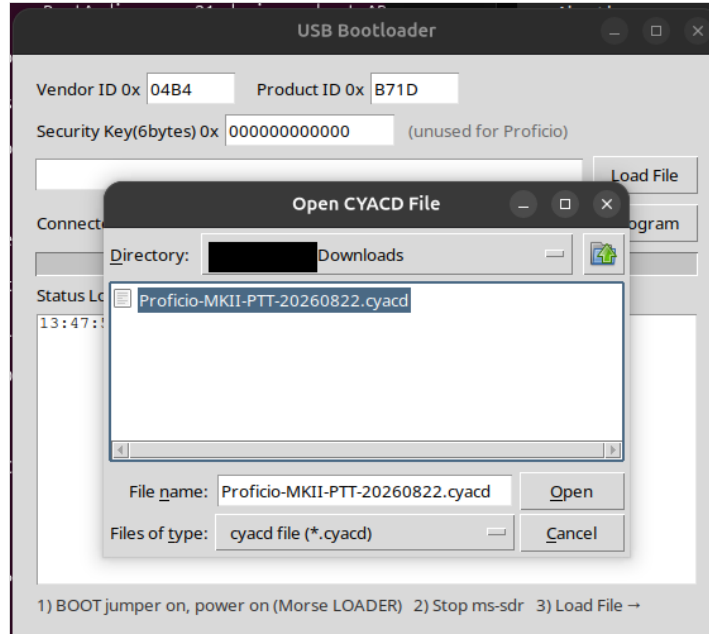


*MSCC Firmware Upload (USB Bootloader), connected to the radio in boot mode*

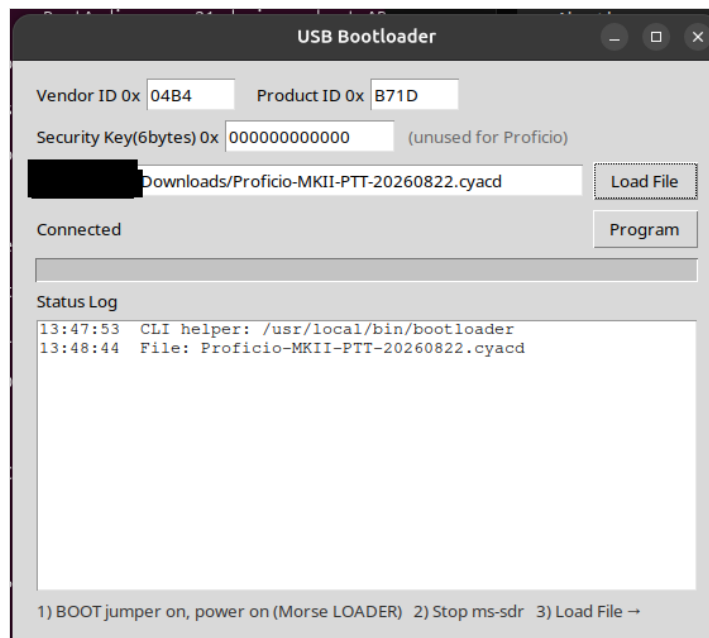
- **Vendor ID (04B4) and Product ID (B71D).** Leave as they are.
- **Security Key.** Leave at all zeros (not used for Proficio).
- **File box and Load File.** Choose the firmware file.
- **Connected.** Shown when the radio is in boot mode and plugged in.
- **Program.** Starts the update.
- **Progress bar and Status Log.** Progress and the result.

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7. **Choose the firmware file.** Click **Load File**. In the **Open CYACD File** window, go to the folder where you saved the file (Downloads in the example), pick the .cyacd file for your radio, and click **Open**. The file's full path now shows in the box next to Load File.



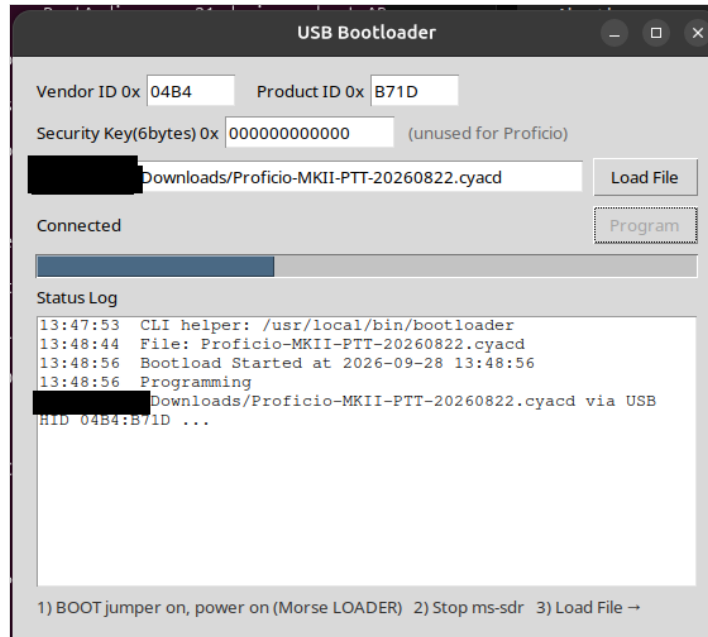
*Open CYACD File: pick the .cyacd file for your radio*



*File loaded; the path shows next to Load File*

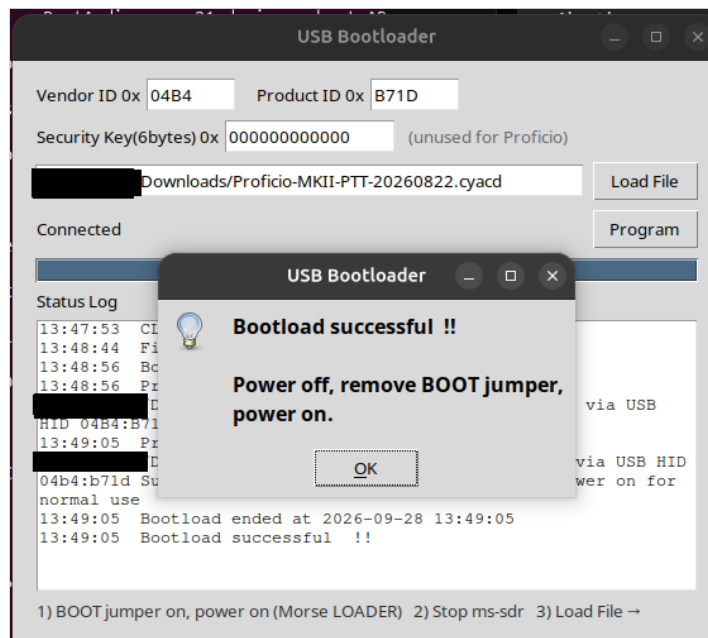
8. **Click Program.** The progress bar fills in a few seconds. **Don't interrupt the update.**

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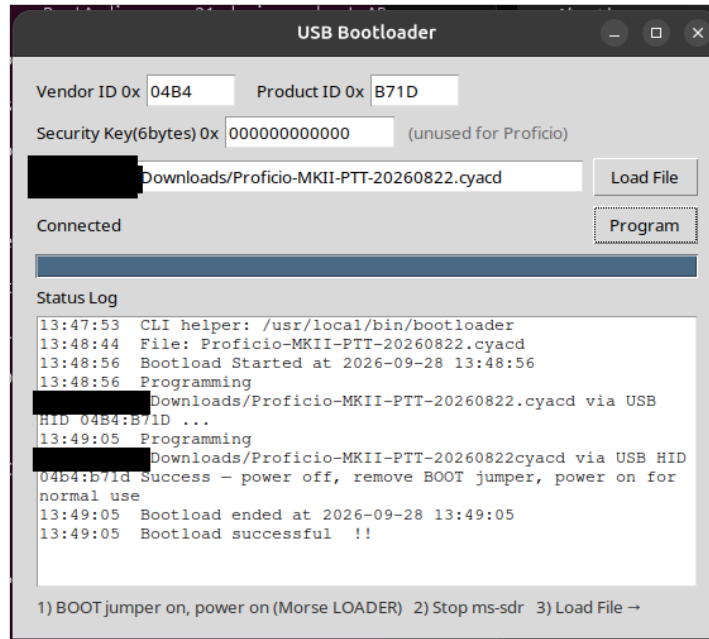
*Programming in progress*

9. Check for the success message. A box says “**Bootload successful !! Power off, remove BOOT jumper, power on.**” Click OK. The Status Log also ends with **Bootload successful !!**



*Bootload successful !! Power off, remove BOOT jumper, power on.*

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*Update finished: the Status Log ends with “Bootload successful !!”*

10. **Take the shunt off the boot pins.** Power the radio **off** and remove the shunt from the two boot pins. You can leave it off, or park it on a single pin or on the other pins so it doesn't get lost; either is harmless. Then power the radio on again.
11. **Check the version.** Close Firmware Upload, start MSCC UI and connect. The title bar should show the new firmware version (FW).

A Terminal alternative for step 5–9 is `bootloader /path/to/file.cyacd` (with the radio in boot mode).

## □ Appendix C: WSJT-X Setup

WSJT-X talks to the MSCC servers directly through a virtual serial port. MSCC looks like a **Kenwood TS-2000** to WSJT-X. MSCC UI doesn't have to be open for WSJT-X to work, but it is where you see the spectrum and change modes and power.

**Find your port and audio names first.** Open MSCC UI → **SETTINGS** → **Radio host (mscc-init)** (4.10). The line **CAT (WSJT-X)** is the serial port for WSJT-X, and the orange hint line repeats the settings below. Which port it is depends on the port you picked in MSCC Init (2.4, step 3). The ports come in pairs (tnt0 with tnt1), and WSJT-X should normally use the other end of the servers' pair: in the example the servers use /dev/tnt1, so SETTINGS shows /dev/tnt0 (also listed as /dev/ttyUSB10) for WSJT-X. Setups differ (on a Raspberry Pi it may be the other way round), so always use the port SETTINGS shows on your computer.

**Recommended WSJT-X settings** (File → Settings → Radio and Audio)

| WSJT-X setting           | Value                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rig                      | Kenwood TS-2000                                                                                                                                               |
| Serial port              | The <b>CAT (WSJT-X)</b> port from SETTINGS, for example /dev/tnt0. If WSJT-X's list only shows ttyUSB names, pick the "also" name, for example /dev/ttyUSB10. |
| Baud rate                | 9600                                                                                                                                                          |
| Handshake                | None                                                                                                                                                          |
| PTT method               | <b>CAT</b>                                                                                                                                                    |
| Mode                     | <b>None</b> . Then select <b>DIG-U</b> yourself in MSCC UI.                                                                                                   |
| Split operation          | None                                                                                                                                                          |
| Audio Input (Soundcard)  | <b>VirtualB.monitor</b> (Digital Input)                                                                                                                       |
| Audio Output (Soundcard) | <b>VirtualA</b> (Digital Output)                                                                                                                              |
| TX tone                  | Around 1500 Hz                                                                                                                                                |

### Steps

1. In MSCC UI, select **DIG-U**. The audio path switches to Digital.
2. Choose a Hi filter. In DIG-U you can narrow it to 1.4k or 1.0k.
3. **AMP**: leave it off for barefoot 5 W operation. Turn it on only when your amplifier is in line and AMP CAL has been done for that band. **ALC**: keep it on (RX/TX tab).
4. In WSJT-X, click **Test CAT**, then **Test PTT**.
5. Press **Tune** in WSJT-X. Adjust **Digital Mic** (left rail) until the ALC meter is just into the yellow.



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**Only one program on each end of the CAT pair.** The servers use one end (CAT (ms-sdr)) and WSJT-X uses the other (CAT (WSJT-X)). Never point WSJT-X at the servers' end.



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### □ Appendix D: Where Settings and Files Live

The ~ below means your home folder (for example /home/yourname).

| What                                                          | Where                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| MSCC UI settings (window, display, last used, colors)         | ~/.config/MSCC/mscc-avalonia.ini                                                                     |
| Favorites                                                     | ~/.config/MSCC/mscc-favorites.ini                                                                    |
| MSCC UI log                                                   | ~/.local/share/MSCC-Avalonia/logs/mscc.log                                                           |
| Settings written by MSCC Init (CAT port, speaker, mic, keyer) | ~/.local/mscc/ (for example comm-port.ini, operator-speaker.ini, mscc.ini)                           |
| Radio calibration and server logs                             | ~/.local/mscc/ (the same file names as on Windows: iq.ini, recv-iq.ini, power_cal.ini, freq_cal.ini) |
| Server programs                                               | ~/mscc/ (started with the mscc command)                                                              |

Folders whose names start with a dot are hidden. In the Files app press **Ctrl+H** to show them.



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### ☐ Appendix E: Troubleshooting (FAQ)

**“MSCC is not set up on this computer yet.”** Run MSCC Init (2.4). You can click OK and keep going in the meantime.

**Connect does nothing, or the spectrum stays empty.** Check the radio is powered on and its USB cable is plugged in. Open **MSCC Status** and look for FAIL lines. With Launch unchecked, make sure you ran MSCC Start first.

**“WARNING: keep-alive lost”.** MSCC UI stopped hearing from the servers. Disconnect, check the radio’s power and USB, and Connect again. If needed, MSCC Stop then MSCC Start (or `mscc restart`).

**No firmware or core version in the title after connecting.** Check the radio’s USB and power, then Disconnect and Connect again (or restart the servers).

**Launch is grayed out.** Launch works only while disconnected and with Host **127.0.0.1**.

**Remote is grayed out.** That’s normal for local operation (Host 127.0.0.1). Remote is only for a radio on another computer.

**No sound, or the wrong speaker or microphone.** Run MSCC Init again (it stops the servers for you), pick the right devices in step 4, Write config, and start the servers. Check the choices in SETTINGS → Radio host.

**WSJT-X can’t open the serial port.** Use the **CAT (WSJT-X)** port from SETTINGS (Appendix C), not the servers’ port. On Ubuntu, make sure you ran the `usermod` command and logged out and back in (2.2).

**The /dev/tnt ports are gone after a kernel update.** After an Ubuntu or Raspberry Pi OS kernel update, reinstall the servers package (run `./install-mscc.sh` again from the install folder, as in 2.2) so the `ttyOtty` driver is rebuilt for the new kernel and the `/dev/tnt` ports come back.

**WSJT-X keys the radio but power is very low.**

- Move the WSJT-X TX tone to about 1500 Hz.
- Check AMP. AMP off = normal QRP 5 W. Use AMP on only with an amplifier in line and a calibrated AMP table.
- Check that ALC on the RX/TX tab is on.
- Raise Digital Mic until ALC is just into the yellow.

**CW paddle does nothing.** If you use an external keyer or a Legacy radio, check **Use external electronic keyer (legacy)** on the CW tab and restart the servers. For an MKII radio with a paddle, leave it unchecked.



## **MSCC Operator's Guide — Linux and Raspberry Pi (Preliminary)**

**Sending a problem report.** Don't restart before you collect the information: a picture of MSCC Status, the MSCC UI log (`~/.local/share/MSCC-Avalonia/logs/mscc.log`) and the server logs in `~/.local/mscc/`. Post on the forum, <https://multusdr.groups.io/g/main>